

Supplementary Material

A data-adaptive audit of spatial transfer in environmental residual-correction models

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How this Supplement is organised

The main article makes one argument: a residual-correction gain is not a spatial-transfer gain until the dependence pathways connecting the held-out target to its training set have been blocked, and the cost of blocking them has been measured. This Supplement carries the evidence in three parts.

- **Sections S1–S3** document the data, the CAMS extraction, and the estimation and inference protocol.
- **The empirical audit** is carried by Sections S9, S13–S15, S17 and S18, with Tables S6, S7, S10 and S20–S26 and S28–S34: the buffer ladder with the guard band as its contrast, the block-construction and quality-control variants, the coordinate ablation and the paired comparison, the coordinate-permutation placebo, the grid-origin sensitivity, the nearest-neighbour and area-of-applicability analysis, the source-set stability check and the learner sensitivity.
- **The forecasting benchmark** is carried by Sections S4–S8 and S10–S12, with Tables S1–S5, S8, S9, S11–S19 and S27 and Figures S1 and S2. That arm established why a cold start is the case worth auditing at all, but it is not a co-equal contribution of this article and is reported here rather than in the main text.
- **Section S19** reports the ground-truth stochastic experiment in full (Tables S35–S40), including the corner of the factorial in which the audit performs worst.

Sections and tables S1–S34 keep the numbering of the earlier version of this work, so that numbers quoted in the deposited artefacts continue to resolve. Nothing has been renumbered; the stochastic-experiment tables are appended after the last existing table.

S1 Data and quality control

Hourly $\text{PM}_{2.5}$ and PM_{10} observations were retrieved from the OpenAQ open-data archive (public Amazon S3 mirror; no key required). Sentinel (-999) and negative values were removed; no upper concentration cap was applied, so the record’s largest PM_{10} value ($2,978 \mu\text{g}/\text{m}^3$ at E11 Road on 2023-09-15) is retained, and records were averaged onto the three-hourly CAMS grid. Table S1 lists all stations.

S2 CAMS extraction

CAMS global reanalysis (EAC4) surface $\text{PM}_{2.5}$ (`pm2p5`) and PM_{10} (`pm10`) were retrieved at 0.75° , three-hourly, and converted from kg/m^3 to $\mu\text{g}/\text{m}^3$ ($\times 10^9$). The grid cell nearest each station was used.

S3 Models and evaluation

The Chronos family (Chronos-2 and Chronos-Bolt) was evaluated across all 24 site–pollutant series; an independent-family model (TimesFM) was also run as a zero-shot spot-check at six sites (Table S11). Models were applied zero-shot with context length K (default 64 steps) and horizon $H = 8$ (24 h). Rolling-origin evaluation used up to 120 equally spaced origins per station; each origin’s context window spans $[t - K, t - 1]$ and its targets span $[t, t + H - 1]$, so the target never enters the context (no look-ahead leakage). After quality control this yields 4,800 (PM_{2.5}) and 9,920 (PM₁₀) evaluated forecast points overall, of which 230 and 446 fall in the top-5% extreme subset. Within-station samples are strongly autocorrelated, so the station is the coarsest defensible resampling unit for the benchmark arm, and 95% confidence intervals there were obtained by a station-level cluster bootstrap (2000 resamples of stations with replacement), which keeps each point estimate within its interval. Paired model differences are summarised by Cohen’s d_z with Wilcoxon signed-rank p -values, reported both raw and under Benjamini–Hochberg and Holm step-down corrections across the ten comparisons (Table S8). We report detection skill as probability of detection (POD), false-alarm ratio (FAR) and critical success index (CSI), in two ways: (i) as *formal WHO 24-h mean* guideline exceedance, where each origin’s eight-step (24-h) forecast mean is compared against the observed 24-h mean (Table S12); and (ii) as instantaneous three-hourly operational concentration tiers, a routine-elevated tier and a severe short-duration dust tier, which are not WHO health metrics (Table S9). Additional error metrics (RMSE, mean bias, normalised MAE, and macro/median MAE) are in Table S13. Extreme events are targets above the station 95th percentile. The cold-start transfer used a histogram gradient-boosting regressor (300 iterations, learning rate 0.05, depth 6) on features {CAMS value, hour sin/cos, day-of-year sin/cos, latitude, longitude}; a coordinate-free ablation removing latitude/longitude is reported in Table S6. Internal early stopping is disabled and `random_state` is fixed at 0, which makes the fit deterministic at this training-set size: repeating it under different seeds reproduces every per-station mean absolute error bit-for-bit. The audit applies three controls jointly. (i) A *distance buffer*: every source station within radius $R \in \{0, 25, 50, 100\}$ km of the target is withheld, and the number of surviving sources is reported. (ii) *Time blocking*: the target’s record is split into six contiguous blocks, taken at quantiles of the target’s own timestamps, and each block is predicted from source data outside that block and outside a guard band set from the measured residual autocorrelation (ten days; a three-day guard chosen to exceed a synoptic dust episode is reported alongside it). (iii) A *leave-one-grid-cell-out* stage: every source sharing the target’s 0.75° CAMS cell is withheld, because such stations carry identical covariate series. A coordinate-free ablation (latitude and longitude removed) is crossed with all of these. Stability is assessed by resampling the source-station set (100 bootstrap source sets per target, Table S20), which is the resampling that carries information for a deterministic estimator; the earlier seed-based variant measured the sensitivity of scikit-learn’s default early-stopping holdout and is not reported. The coordinate-permutation placebo is run under the reported specification at the two radii that bracket the argument, with 99 permutations each, so the smallest attainable one-sided p is 0.010 (Table S24), and the audit is repeated with four learners spanning a wide capacity range (Table S25). All gains carry station cluster-bootstrap 95% intervals (2000 resamples), and tests within the audit family are Benjamini–Hochberg corrected across the twenty-four buffer $\times$ feature-set $\times$ design comparisons of Table S6, with the sign test and the Wilcoxon test corrected as separate families. The family is 24 cells but not 24 distinct statistics: with 19 stations the exact sign test takes only six distinct values across those cells (the Wilcoxon test takes fifteen), because cells sharing a win count share a p value. Benjamini–Hochberg remains valid under such ties and under the positive dependence these cells have, but the correction is correspondingly less costly than a family of 24 independent tests would imply, and we report the uncorrected p values alongside

q throughout so that the reader can apply a different family definition. Two robustness analyses supplement the main benchmark: a like-for-like comparison against the archived CAMS operational forecast on the same PM₁₀ origins (Table S14, Figure S1), a per-country breakdown of the headline result (Table S15), an episode-block bootstrap that resamples whole multi-day dust episodes to check the PM₁₀ significance against event-scale dependence (Table S16), a grid-representativeness analysis (Table S17) with a direct test that the headline survives on the best-represented stations (Table S18), and an independent-model cross-check against NASA MERRA-2 (Table S19).

Table S1: Monitoring-station inventory: pollutant, coordinates, observation period, number of matched three-hourly samples, and mean CAMS bias. Bias and sample counts are computed over each station’s full matched record (for PM₁₀, 2022-10-29 to 2023-12-31); the rolling-origin evaluation uses the September–December 2023 subset of these records.

Station	Pollutant	Lat	Lon	Period	<i>N</i>	CAMS bias (µg/m ³)
Manama	PM2.5	26.20	50.57	2016-12-01–2023-12-22	13745	+6.6
Kuwait City	PM2.5	29.29	48.05	2017-02-17–2023-12-31	15771	+36.9
Dhahran	PM2.5	26.28	50.15	2019-10-28–2021-07-14	3545	+11.5
Jeddah	PM2.5	21.54	39.17	2020-06-15–2023-12-31	3095	+3.8
Doha	PM2.5	25.33	51.23	2022-04-14–2023-11-30	2574	+15.1
Bida Zayed	PM10	23.65	53.70	2022-10-29–2023-12-31	1073	-5.5
Khadeeja School	PM10	24.48	54.37	2022-10-29–2023-12-31	1127	-60.0
Liwa	PM10	23.10	53.61	2022-10-29–2023-12-31	1083	+8.3
Hamdan St.	PM10	24.49	54.36	2022-10-29–2023-12-31	1081	-53.4
Mussafah	PM10	24.35	54.50	2022-10-29–2023-12-31	1080	-73.0
Al Ain St.	PM10	24.23	55.77	2022-10-29–2023-12-31	1068	-1.4
Sweihan	PM10	24.47	55.34	2022-10-29–2023-12-31	1099	+37.3
Khalifa City	PM10	24.42	54.58	2022-10-29–2023-12-31	1079	-52.4
Gayathi	PM10	23.84	52.81	2022-10-29–2023-12-31	1144	-19.7
Al Maqtaa	PM10	24.40	54.52	2022-10-29–2023-12-31	1083	-58.9
Ruwais	PM10	24.09	52.75	2022-10-29–2023-12-31	1134	-3.4
Baniyas School	PM10	24.32	54.64	2022-10-29–2023-12-31	1108	-35.6
Habshan S.	PM10	23.75	53.75	2022-10-29–2023-12-31	1076	-14.1
Al Ain Islamic	PM10	24.22	55.73	2022-10-29–2023-12-31	1108	+2.5
Zakher	PM10	24.16	55.70	2022-10-29–2023-12-31	1134	+10.7
Al Qua'a	PM10	23.53	55.49	2022-10-29–2023-12-31	1094	-8.2
Al Tawia	PM10	24.26	55.70	2022-10-29–2023-12-31	1096	-0.8
E11 Road	PM10	24.04	53.89	2022-10-29–2023-12-31	1067	-73.9
Al Mafraq	PM10	24.29	54.59	2022-10-29–2023-12-31	1084	-121.2

S4 Operational-forecast comparison (PM₁₀)

To test whether our use of the EAC4 reanalysis rather than a real-time product drives the conclusion, we repeated the PM₁₀ benchmark with the archived CAMS *global atmospheric-composition operational forecast* (`cams-global-atmospheric-composition-forecasts`, initialised 00 and 12 UTC, particulate_matter_10um), retrieved from the Atmosphere Data Store and treated identically to the reanalysis ($\times 10^9$ conversion, nearest grid cell). For each rolling origin t_0 the operational nowcast uses the most recent 00/12 UTC forecast cycle at or before t_0 , and the prediction for each target valid time is that cycle’s value at the corresponding lead (0–33 h); this ignores real-time dissemination latency and is therefore favourable to CAMS. The operational forecast was sampled on the *same* origins and targets as the main PM₁₀ benchmark: the reanalysis point estimates are reproduced identically on this matched set (9,920 forecast points; raw-CAMS MAE 43.0, 24-h-mean POD 0.72), so the operational forecast is added as a strictly like-for-like column (persistence, which requires a defined immediately preceding observation, is scored on the 9,480 of these points where one exists). The small CI differences between the main benchmark table (Table S2) and this matched operational-forecast table reflect independent station-bootstrap draws around the same point estimate, not different underlying forecast targets; the machine-readable supplement records both table outputs and seeds. All PM₁₀ origins fall in the September–December 2023 dust period, which the operational archive covers in full; the PM_{2.5} operational forecast (whose evaluation window extends back to 2016, before the archive) was not reconstructed and is left for future work.

Table S2: Benchmark MAE ($\mu\text{g}/\text{m}^3$) with bootstrap 95% confidence intervals, all events and extreme (top 5%) events.

Model	MAE	95% CI
<i>PM_{2.5}: all events</i>		
raw CAMS	26.4	[17.8, 37.7]
bias-corrected CAMS	23.1	[16.7, 29.7]
persistence	16.5	[12.8, 20.3]
seasonal-naive	18.9	[14.0, 24.0]
FM (Chronos-Bolt)	14.8	[11.2, 19.1]
FM (Chronos-2)	14.8	[11.4, 18.9]
<i>PM_{2.5}: extreme (top 5%)</i>		
raw CAMS	68.3	[38.2, 101.3]
bias-corrected CAMS	76.8	[39.4, 117.7]
persistence	89.5	[47.3, 135.4]
seasonal-naive	105.1	[56.7, 147.1]
FM (Chronos-Bolt)	90.8	[54.6, 134.3]
FM (Chronos-2)	91.2	[54.4, 134.0]
<i>PM₁₀: all events</i>		
raw CAMS	43.0	[33.3, 55.4]
bias-corrected CAMS	30.3	[25.3, 37.1]
persistence	13.0	[9.8, 17.4]
seasonal-naive	22.0	[16.2, 30.3]
FM (Chronos-Bolt)	14.7	[10.7, 20.7]
FM (Chronos-2)	13.3	[10.2, 17.6]
<i>PM₁₀: extreme (top 5%)</i>		
raw CAMS	107.8	[63.2, 176.2]
bias-corrected CAMS	75.3	[46.6, 121.2]
persistence	42.1	[27.8, 63.5]
seasonal-naive	79.2	[46.8, 128.7]
FM (Chronos-Bolt)	47.6	[28.5, 77.1]
FM (Chronos-2)	40.9	[27.1, 60.9]

Table S3: MAE ($\mu\text{g}/\text{m}^3$) by lead time for raw CAMS and the foundation model (Chronos-2).

Lead (h)	PM _{2.5}		PM ₁₀	
	CAMS	FM	CAMS	FM
3	25.7	9.4	44.2	3.6
6	25.5	11.7	45.0	6.7
9	25.8	15.1	43.3	9.4
12	25.6	14.9	44.3	12.2
15	27.3	16.3	41.7	15.0
18	27.4	16.8	42.3	17.5
21	27.3	17.7	40.7	19.9
24	26.2	16.2	42.5	21.6

Table S4: Foundation-model and CAMS MAE ($\mu\text{g}/\text{m}^3$) and their ratio by observed-concentration percentile bin. Ratio > 1 means CAMS is more accurate. Bins are percentiles of the pooled observed-concentration distribution and therefore do not coincide with the station-wise extreme subset of Table S2, which is defined by each station’s own 95th percentile; the two stratifications are not expected to agree numerically and the crossover statement in the main text is supported by both independently.

Pollutant	Percentile	FM MAE	CAMS MAE	FM/CAMS
PM _{2.5}	0–25	8.9	21.2	0.42
PM _{2.5}	25–50	7.8	26.2	0.30
PM _{2.5}	50–75	9.6	25.9	0.37
PM _{2.5}	75–90	14.0	25.3	0.55
PM _{2.5}	90–95	21.9	31.8	0.69
PM _{2.5}	95–99	41.1	34.7	1.18
PM _{2.5}	99–100	147.1	100.1	1.47
PM ₁₀	0–25	7.0	25.3	0.28
PM ₁₀	25–50	8.5	26.3	0.32
PM ₁₀	50–75	10.6	39.0	0.27
PM ₁₀	75–90	17.6	54.6	0.32
PM ₁₀	90–95	20.6	75.9	0.27
PM ₁₀	95–99	45.8	118.1	0.39
PM ₁₀	99–100	108.3	336.3	0.32

Table S5: Chronos-2 with CAMS as a covariate (PM_{2.5}, five sites, dedicated covariate-sweep origin subset; the raw-CAMS reference therefore differs slightly from the pooled Table S2 value). Mean MAE ($\mu\text{g}/\text{m}^3$) across stations by context length K : raw CAMS, foundation model on observations (c2_pure), and with the CAMS covariate (c2_cov). The covariate helps only marginally at the longest context ($K = 32$) and hurts at short context.

K	raw CAMS	FM (obs)	FM + CAMS covariate
2	25.3	16.4	18.7
4	25.3	18.5	18.7
8	25.3	17.4	18.2
16	25.3	14.9	15.6
32	25.3	14.6	14.4

Table S6: **The audit ladder, with the guard band as the contrast.** The primary quantity is $\Delta = \log(\text{MAE}_{\text{raw}}/\text{MAE}_{\text{transfer}})$, declared in advance; $\Delta = +0.13$ is a 12% error reduction. Everything is held fixed across the two halves of the table except the guard band separating each time block from its training data: three days, chosen to exceed a synoptic dust episode, against ten days, the lag at which the residual autocorrelation of this network reaches +0.08 (Table S30). Time blocks are quantiles of each target’s own timestamps, so the six folds hold 170 to 190 points rather than the 4 to 517 that calendar-equal blocks produce on a record with empty months. The flat-line quality control of Table S31 is applied throughout. Intervals are a block bootstrap over the nine 0.75° CAMS cells (2000 resamples), which is the inference unit we adopt because 16 of the 19 stations share a cell and every same-cell pair carries an identical covariate series. q is the Benjamini–Hochberg-corrected exact sign test across the twenty-four cells of each half. Win counts are for the ten-day guard. Under the three-day guard both feature sets clear correction at the two narrowest radii; under the ten-day guard nothing clears it anywhere, at a cost of 11.6% of the training data.

R (km)	Sources	guard = 3 days			guard = 10 days			Win
		Median Δ	Cell 95% CI	q	Median Δ	Cell 95% CI	q	
<i>distance buffer only: with lat/lon</i>								
0	18.0	+0.256	[+0.157, +0.467]	0.008	+0.256	[+0.157, +0.467]	0.027	16/19
25	15.5	+0.105	[-0.017, +0.192]	0.706	+0.105	[-0.017, +0.192]	0.777	11/19
50	14.4	-0.017	[-0.104, +0.182]	1.000	-0.017	[-0.104, +0.182]	1.000	9/19
100	12.0	-0.022	[-1.042, +0.275]	1.000	-0.022	[-1.042, +0.275]	1.000	9/19
<i>distance buffer only: coordinate-free</i>								
0	18.0	+0.362	[+0.120, +0.441]	0.008	+0.362	[+0.120, +0.441]	0.027	16/19
25	15.5	+0.219	[+0.120, +0.396]	0.008	+0.219	[+0.120, +0.396]	0.027	16/19
50	14.4	+0.182	[+0.070, +0.321]	0.085	+0.182	[+0.070, +0.321]	0.251	14/19
100	12.0	+0.139	[-0.100, +0.321]	0.706	+0.139	[-0.100, +0.321]	0.777	11/19
<i>buffer + time blocks: with lat/lon</i>								
0	18.0	+0.251	[+0.158, +0.337]	0.008	+0.207	[+0.106, +0.302]	0.017	17/19
25	15.5	+0.143	[+0.072, +0.218]	0.006	+0.045	[-0.086, +0.161]	0.251	13/19
50	14.4	+0.074	[-0.201, +0.143]	0.085	+0.073	[-0.362, +0.191]	0.251	13/19
100	12.0	+0.057	[-0.924, +0.249]	0.706	+0.034	[-0.491, +0.191]	1.000	10/19
<i>buffer + time blocks: coordinate-free</i>								
0	18.0	+0.181	[+0.025, +0.335]	0.008	+0.220	[-0.023, +0.254]	0.251	13/19
25	15.5	+0.161	[+0.027, +0.265]	0.008	+0.147	[-0.061, +0.221]	0.251	13/19
50	14.4	+0.161	[+0.005, +0.260]	0.008	+0.127	[-0.080, +0.229]	0.251	13/19
100	12.0	+0.072	[-0.025, +0.281]	0.085	+0.087	[-0.162, +0.244]	0.777	11/19
<i>buffer + time + leave-one-grid-cell-out: with lat/lon</i>								
0	16.4	+0.192	[+0.059, +0.317]	0.006	+0.132	[-0.015, +0.228]	0.251	13/19
25	15.2	+0.143	[+0.056, +0.218]	0.006	+0.073	[+0.016, +0.161]	0.251	14/19
50	14.4	+0.074	[-0.201, +0.160]	0.085	+0.073	[-0.362, +0.191]	0.251	13/19
100	12.0	+0.057	[-0.924, +0.249]	0.706	+0.034	[-0.618, +0.191]	1.000	10/19
<i>buffer + time + leave-one-grid-cell-out: coordinate-free</i>								
0	16.4	+0.181	[+0.005, +0.325]	0.008	+0.218	[-0.061, +0.232]	0.251	13/19
25	15.2	+0.161	[+0.005, +0.262]	0.008	+0.147	[-0.041, +0.221]	0.251	13/19
50	14.4	+0.161	[+0.016, +0.260]	0.008	+0.127	[-0.041, +0.229]	0.251	13/19
100	12.0	+0.072	[+0.000, +0.281]	0.085	+0.087	[-0.162, +0.244]	0.777	11/19

Table S7: Per-station zero-observation cold-start transfer for the PM₁₀ network (leave-one-station-out, all other 18 stations as sources): observed and CAMS means, CAMS bias, the observation–CAMS temporal correlation, and transfer gain over raw CAMS (%). Negative gains mark over-correction; at Ruwais ($-3.4 \mu\text{g m}^{-3}$ bias) this is over-correction of an already accurate field, whereas Baniyas School ($-35.6 \mu\text{g m}^{-3}$) is a moderately biased site at which transfer nonetheless fails. Across the 19 stations the gain is uncorrelated with $|\text{bias}|$ (Spearman -0.13 , Pearson $+0.18$; -0.23 excluding Ruwais), so target-site bias does not predict transfer benefit. Station-bootstrap 95% intervals (2,000 resamples of the 19 stations) are $[+13.9, +32.5]\%$ for the median gain and $[-28.0, +25.9]\%$ for the mean: the median excludes zero, the mean does not, because percentage gain is bounded above by 100% and unbounded below and is therefore dominated by the single over-corrected station.

Station	Obs mean	CAMS mean	Bias	Corr.	Transfer gain (%)
Al Mafrq	176	55	-121.2	0.37	+26.8
E11 Road	125	51	-73.9	0.39	+13.9
Mussafah	128	55	-73.0	0.12	+39.9
Khadeeja Sch.	103	43	-60.0	0.38	+24.3
Al Maqtaa	109	50	-58.9	0.33	+32.5
Hamdan St.	96	43	-53.4	0.11	+9.8
Khalifa City	103	50	-52.4	0.39	+15.6
Baniyas Sch.	90	55	-35.6	0.50	-61.0
Gayathi	79	60	-19.7	0.52	+15.4
Habshan S.	65	51	-14.1	0.40	+17.0
Al Qua'a	93	84	-8.2	0.19	+14.2
Bida Zayed	56	51	-5.5	0.31	+6.7
Ruwais	63	59	-3.4	0.36	-244.1
Al Ain St.	83	82	-1.4	0.50	+46.4
Al Tawia	83	82	-0.8	0.50	+44.2
Al Ain Islamic	79	82	+2.5	0.30	+44.0
Liwa	71	80	+8.3	0.29	+14.1
Zakher	72	82	+10.7	0.43	+37.9
Sweiha	68	106	+37.3	0.38	-7.9

Table S8: Paired effect sizes for key model comparisons (unit: station); positive Cohen’s d_z favours the first model. p is a two-sided Wilcoxon signed-rank test; p_{BH} and p_{Holm} are Benjamini–Hochberg and Holm corrections across all ten comparisons. **The five PM_{2.5} comparisons are underpowered by design:** with $n = 5$ stations the smallest attainable two-sided Wilcoxon p is 0.0625, so no PM_{2.5} comparison can reach significance; these are reported as descriptive effect sizes only (see main text). The PM₁₀ gains over CAMS survive both corrections. The FM↔persistence comparison is non-significant ($d_z = -0.12$, $p = 0.52$), which we report as unresolved rather than as evidence of equivalence: the 95% interval on d_z (approximately $[-0.60, +0.36]$) excludes neither a moderate foundation-model advantage nor a moderate persistence advantage, and the ordering reverses under RMSE (Table S13).

Pollutant	Comparison	n	d_z	p	p_{BH}	p_{Holm}
PM _{2.5}	FM (Chronos-2) vs raw CAMS	5	+1.37	0.062	0.069	0.375
PM _{2.5}	FM (Chronos-Bolt) vs raw CAMS	5	+1.39	0.062	0.069	0.375
PM _{2.5}	FM (Chronos-2) vs persistence	5	+2.06	0.062	0.069	0.375
PM _{2.5}	FM (Chronos-2) vs bias-corr. CAMS	5	+1.78	0.062	0.069	0.375
PM _{2.5}	persistence vs raw CAMS	5	+1.13	0.062	0.069	0.375
PM ₁₀	FM (Chronos-2) vs raw CAMS	19	+1.49	<0.001	<0.001	<0.001
PM ₁₀	FM (Chronos-Bolt) vs raw CAMS	19	+1.38	<0.001	<0.001	<0.001
PM ₁₀	FM (Chronos-2) vs persistence	19	-0.12	0.515	0.515	0.515
PM ₁₀	FM (Chronos-2) vs bias-corr. CAMS	19	+1.75	<0.001	<0.001	<0.001
PM ₁₀	persistence vs raw CAMS	19	+1.53	<0.001	<0.001	<0.001

The operational forecast is statistically indistinguishable from the reanalysis on error and, like it, roughly three times the observation-driven nowcast error (Table S14, Figure S1). On WHO 24-h mean detection it flags more exceedances than the reanalysis (POD 0.92 vs. 0.72) but only by over-forecasting dust, so its balanced skill (CSI 0.86) stays below the observation-driven nowcasts (CSI 0.94–0.95). The choice of CAMS product therefore does not change the routine-nowcast regime map: on error the two CAMS products are indistinguishable, and although the operational forecast scores higher on 24-h detection (CSI 0.86 vs. the reanalysis 0.68) it does so by over-forecasting dust and still trails the observation-driven nowcasts. This like-for-like comparison does not evaluate first-onset lead time; that remains a separate prospective forecast-verification question.

S5 Cross-region robustness

The observation-driven advantage is present in every national network sampled, not only in aggregate (Table S15). For PM_{2.5} the five sites span four countries and both the Gulf and Red Sea coasts; Chronos-2 lowers MAE over the CAMS reanalysis at every one, by 20% (Jeddah, Red Sea, the mildest site) to 54% (Dhahran). Each PM_{2.5} country is represented by a single station, so these are per-station point estimates rather than cluster-bootstrap intervals; the PM₁₀ row is the 19-station Abu Dhabi network with intervals. A fully external region and season (outside the Arabian Peninsula) remains the strongest outstanding generalisation test.

S6 Sensitivity to temporal (dust-event) dependence

The PM₁₀ confirmatory arm is a single dense cluster (median nearest-neighbour distance 6.5 km) observed in one late-2023 (September–December) dust period: all PM₁₀ origins fall between 1 September and 30 December 2023, and because dust is synoptic, the truly independent sampling units are

Table S9: Detection skill for *instantaneous three-hourly* concentration tiers (operational alert proxies, *not* formal WHO 24-h metrics; the WHO 24-h mean analysis is Table S12): a routine-elevated tier and a severe short-duration tier. POD and CSI with station cluster-bootstrap 95% intervals, and FAR. The routine-elevated tiers have high base rates (PM_{2.5} 88.6%, PM₁₀ 90.0% of evaluated points), so CSI values near 0.9 partly reflect the base rate rather than skill; the severe tiers are far rarer and more discriminating. The severe short-duration dust tier ($\geq 150 \mu\text{g}/\text{m}^3$, 3-h) is an operational episode threshold, not a health guideline. Raw CAMS detects almost none of the severe short-duration dust episodes, whereas observation-driven nowcasts recover roughly two-thirds.

Model	POD [95% CI]	FAR	CSI [95% CI]
<i>PM_{2.5}: routine-elevated 3-h tier ($\geq 15 \mu\text{g}/\text{m}^3$, 4251 events)</i>			
raw CAMS	0.94 [0.87, 0.98]	0.08	0.86 [0.74, 0.93]
bias-corrected CAMS	0.82 [0.78, 0.86]	0.06	0.78 [0.70, 0.83]
persistence	0.93 [0.85, 0.99]	0.06	0.88 [0.76, 0.96]
FM (Chronos-Bolt)	0.95 [0.86, 0.99]	0.06	0.90 [0.77, 0.97]
FM (Chronos-2)	0.95 [0.86, 0.99]	0.06	0.89 [0.78, 0.97]
<i>PM_{2.5}: severe short-duration 3-h tier ($\geq 75 \mu\text{g}/\text{m}^3$, 324 events)</i>			
raw CAMS	0.61 [0.52, 0.72]	0.80	0.18 [0.13, 0.23]
bias-corrected CAMS	0.45 [0.42, 0.51]	0.73	0.21 [0.17, 0.23]
persistence	0.32 [0.22, 0.37]	0.66	0.20 [0.13, 0.22]
FM (Chronos-Bolt)	0.25 [0.16, 0.32]	0.53	0.20 [0.12, 0.23]
FM (Chronos-2)	0.27 [0.14, 0.36]	0.54	0.20 [0.10, 0.27]
<i>PM₁₀: routine-elevated 3-h tier ($\geq 45 \mu\text{g}/\text{m}^3$, 8924 events)</i>			
raw CAMS	0.67 [0.57, 0.76]	0.07	0.63 [0.55, 0.71]
bias-corrected CAMS	0.87 [0.82, 0.92]	0.05	0.83 [0.78, 0.89]
persistence	0.97 [0.95, 0.99]	0.03	0.95 [0.92, 0.97]
FM (Chronos-Bolt)	0.96 [0.94, 0.98]	0.03	0.94 [0.90, 0.96]
FM (Chronos-2)	0.97 [0.95, 0.99]	0.03	0.94 [0.92, 0.97]
<i>PM₁₀: severe short-duration 3-h tier ($\geq 150 \mu\text{g}/\text{m}^3$, 765 events)</i>			
raw CAMS	0.06 [0.03, 0.12]	0.85	0.04 [0.03, 0.07]
bias-corrected CAMS	0.50 [0.24, 0.68]	0.66	0.26 [0.09, 0.41]
persistence	0.68 [0.51, 0.76]	0.36	0.49 [0.34, 0.57]
FM (Chronos-Bolt)	0.68 [0.50, 0.76]	0.33	0.51 [0.33, 0.62]
FM (Chronos-2)	0.67 [0.49, 0.76]	0.36	0.48 [0.30, 0.60]

Table S10: Pooled zero-observation cold-start MAE ($\mu\text{g}/\text{m}^3$) with station cluster-bootstrap 95% intervals: raw CAMS, a global-bias transfer (single mean residual added), and the gradient-boosted residual transfer. Pooling is over all forecast points (station-size weighted), consistent with the main benchmark. Unlike Table S2, the cold-start evaluation uses the full matched record at each held-out station rather than the 120-origin rolling subset, so its raw-CAMS reference values differ from the main benchmark. The transferred residual lowers the pooled error by 35.8% (PM_{2.5}) and 12.0% (PM₁₀). The global-bias transfer (a single source-mean residual added everywhere) does not help for PM_{2.5}, confirming that the useful part of the correction is spatially structured rather than a constant offset.

Pollutant	Model	MAE	95% CI
PM _{2.5}	raw CAMS	33.4	[19.4, 43.7]
PM _{2.5}	global-bias transfer	34.0	[21.8, 39.1]
PM _{2.5}	GBM residual transfer	21.4	[16.6, 24.5]
PM ₁₀	raw CAMS	46.6	[36.4, 58.1]
PM ₁₀	global-bias transfer	45.4	[37.6, 54.5]
PM ₁₀	GBM residual transfer	41.0	[32.0, 51.4]

Table S11: Independent-family cross-check. TimesFM (Google, decoder-only) was run zero-shot as a spot-check at six sites (three $\text{PM}_{2.5}$, three PM_{10}) using the same rolling-origin protocol; N is the number of evaluated origins. TimesFM reproduces the large raw-CAMS error gap seen for the Chronos family, confirming that the observation-driven advantage is not specific to one model family. Because this spot-check uses its own ≤ 60 -origin subset, the per-site raw-CAMS values differ from the full-benchmark values in Table S15.

Site	N	raw CAMS MAE	TimesFM MAE	Reduction (%)
Kuwait City ($\text{PM}_{2.5}$)	60	43.5	17.8	59
Jeddah ($\text{PM}_{2.5}$)	60	13.3	10.0	25
Manama ($\text{PM}_{2.5}$)	60	30.1	18.6	38
Al Mafraq (PM_{10})	56	134.0	39.8	70
Mussafah (PM_{10})	59	64.5	19.1	70
Al Ain St. (PM_{10})	47	29.1	8.2	72

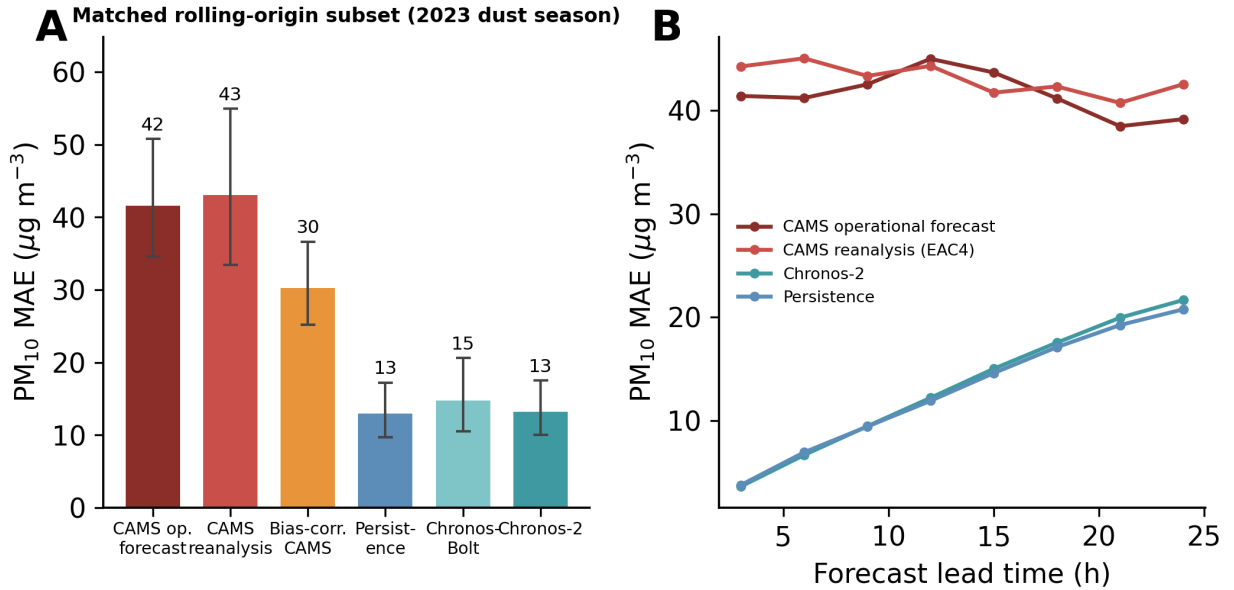


Figure S1: **The CAMS operational forecast behaves like the reanalysis (PM_{10}).** (a) MAE on the matched late-2023 (September–December) dust-period subset with station cluster-bootstrap 95% intervals: the operational forecast (42) and the EAC4 reanalysis (43) are statistically indistinguishable, both roughly three times the observation-driven nowcast error (~ 13). (b) MAE by forecast lead time: both CAMS products stay flat near $40\text{--}45 \mu\text{g m}^{-3}$ across the 24-h horizon (they carry no recent-observation information), whereas the observation-driven nowcasts rise from ~ 4 to $\sim 21 \mu\text{g m}^{-3}$ as the informativeness of the last observation fades.

Table S12: **Proper WHO 24-h mean guideline exceedance detection.** Each forecast origin’s eight-step (24-h) forecast mean is the prospective 24-h average; the observed 24-h mean is the mean of the eight observed steps. Exceedance is evaluated against the WHO 24-h guideline *value* (15 and 45 $\mu\text{g}/\text{m}^3$), so these, unlike Table S9, are genuine 24-h-mean metrics. POD and CSI carry station cluster-bootstrap 95% intervals. Because both guidelines are exceeded at a high base rate ($545/600 = 90.8\%$ for $\text{PM}_{2.5}$, $1119/1240 = 90.2\%$ for PM_{10}), an always-alarm rule already attains CSI 0.91 and 0.90 respectively; chance-corrected scores derived from the same contingency tables are therefore given as equitable threat score (ETS) and Heidke skill score (HSS). Chance-corrected scores now carry the same station cluster-bootstrap intervals (2000 resamples) as POD and CSI, computed by rebuilding the contingency table from the drawn stations. For PM_{10} (19 stations), ETS/HSS with 95% intervals: raw CAMS 0.058 $[-0.006, 0.161]$ / 0.110 $[-0.012, 0.277]$; bias-corrected CAMS 0.269 $[0.130, 0.426]$ / 0.424 $[0.231, 0.597]$; persistence 0.578 $[0.449, 0.683]$ / 0.733 $[0.620, 0.811]$; seasonal-naive 0.452 $[0.310, 0.572]$ / 0.623 $[0.473, 0.728]$. For $\text{PM}_{2.5}$ (5 stations): raw CAMS 0.203 $[-0.010, 0.223]$ / 0.337 $[-0.021, 0.364]$; bias-corrected CAMS 0.236 $[0.037, 0.336]$ / 0.382 $[0.071, 0.503]$; persistence 0.358 $[0.225, 0.407]$ / 0.527 $[0.368, 0.579]$; seasonal-naive 0.436 $[0.178, 0.460]$ / 0.607 $[0.302, 0.630]$. The raw-CAMS interval includes zero for both pollutants, which is a stronger statement than the point estimate alone: on this test the reanalysis is not distinguishable from a chance forecast. The two Chronos rows keep point estimates (0.53/0.69 and 0.54/0.70 for PM_{10} , 0.42/0.59 and 0.45/0.62 for $\text{PM}_{2.5}$) because their per-origin outputs were archived as absolute errors rather than signed predictions, and an absolute error does not determine which side of a threshold a forecast fell on; recomputing their intervals requires re-running the foundation models. Row sample sizes differ where a reference is undefined at some origins (persistence $N = 1185$ for PM_{10} , others $N = 1240$), so chance-corrected scores are computed on each method’s own valid set. ETS and HSS follow Schaefer (1990, *Weather and Forecasting* 5, 570–575). Raw CAMS thus has almost no chance-corrected skill on the PM_{10} test, and on the saturated $\text{PM}_{2.5}$ test neither raw CAMS nor persistence exceeds the always-alarm reference. For PM_{10} the exceedance base rate is high (1119/1240 origin-level 24-h means, 90.2%); an always-alarm reference would therefore have POD 1.00, FAR 0.10 and CSI 0.90. Against this high-base-rate reference, the mechanistic field detects only about seven of ten 24-h exceedances (POD 0.72, CSI 0.68) versus ~ 0.97 for observation-driven nowcasts; for $\text{PM}_{2.5}$ the guideline is exceeded almost ubiquitously and does not discriminate between methods.

Model	POD [95% CI]	FAR	CSI [95% CI]
<i>$\text{PM}_{2.5}$: WHO 24-h mean guideline ($\geq 15 \mu\text{g}/\text{m}^3$, 545 events)</i>			
raw CAMS	0.96 [0.91, 0.99]	0.07	0.90 [0.77, 0.97]
bias-corr. CAMS	0.87 [0.84, 0.89]	0.03	0.84 [0.77, 0.88]
persistence	0.91 [0.82, 0.97]	0.03	0.89 [0.76, 0.97]
seasonal-naive	0.97 [0.90, 0.99]	0.04	0.93 [0.81, 0.99]
FM (Chronos-Bolt)	0.96 [0.88, 0.99]	0.04	0.93 [0.81, 0.99]
FM (Chronos-2)	0.95 [0.87, 0.99]	0.03	0.93 [0.81, 0.99]
<i>PM_{10}: WHO 24-h mean guideline ($\geq 45 \mu\text{g}/\text{m}^3$, 1119 events)</i>			
always-alarm reference	1.00	0.10	0.90
raw CAMS	0.72 [0.61, 0.83]	0.07	0.68 [0.58, 0.78]
bias-corr. CAMS	0.92 [0.89, 0.95]	0.05	0.88 [0.82, 0.93]
persistence	0.96 [0.94, 0.98]	0.02	0.94 [0.91, 0.97]
seasonal-naive	0.97 [0.95, 0.98]	0.04	0.93 [0.90, 0.96]
FM (Chronos-Bolt)	0.96 [0.93, 0.98]	0.02	0.94 [0.90, 0.96]
FM (Chronos-2)	0.97 [0.95, 0.99]	0.03	0.94 [0.91, 0.97]

Table S13: Additional error metrics (all events), reported to complement MAE. RMSE and mean bias expose dust-peak behaviour (raw CAMS under-predicts PM_{10} with mean bias $-23.6 \mu\text{g}/\text{m}^3$ and RMSE 69.6, versus near-zero bias and much lower RMSE for observation-driven methods). Normalised MAE (nMAE) is MAE divided by the mean observed concentration. Because station concentrations differ widely, we report both the station-size-weighted pooled MAE and the un-weighted station-mean (macro) and median MAE.

Model	MAE	RMSE	Bias	nMAE (%)	Macro-MAE	Median-MAE
<i>PM_{2.5}</i>						
raw CAMS	26.4	43.5	+13.4	66.9	26.4	24.8
bias-corr. CAMS	23.1	40.3	-0.8	58.5	23.1	23.3
persistence	16.5	37.8	-1.7	41.8	16.5	15.9
FM (Chronos-Bolt)	14.8	35.6	-3.8	37.6	14.8	13.2
FM (Chronos-2)	14.8	35.7	-3.8	37.5	14.8	13.7
<i>PM₁₀</i>						
raw CAMS	43.0	69.6	-23.6	48.8	43.5	33.8
bias-corr. CAMS	30.3	57.6	+3.4	34.3	31.3	27.7
persistence	13.0	43.0	+2.0	14.7	13.7	10.1
FM (Chronos-Bolt)	14.7	54.2	+2.8	16.7	15.5	10.4
FM (Chronos-2)	13.3	41.5	+3.1	15.0	13.9	10.4

Table S14: **Operational-forecast comparison (PM_{10} , matched late-2023 (September–December) dust-period subset, 9,920 forecast points).** MAE ($\mu\text{g}/\text{m}^3$) with station cluster-bootstrap 95% intervals, and WHO 24-h mean guideline ($\geq 45 \mu\text{g m}^{-3}$) detection. The CAMS operational forecast is sampled on the same rolling origins as the main benchmark (which reproduces the reanalysis point estimates to the reported precision). The operational forecast and the reanalysis are indistinguishable on error and both far worse than the observation-driven nowcasts. The reanalysis CI differs slightly from Table S2 because the operational-forecast comparison was bootstrapped as an independent matched-table run around the same 9,920 targets. *Provenance:* the archived CAMS operational-forecast fields are retrieved from the Copernicus Atmosphere Data Store and are not redistributed in the deposited archive; reproducing this table requires an Atmosphere Data Store account and the request parameters documented in `data_retrieval/RETRIEVAL.md` of the archive; no retrieval client is shipped.

Model	MAE [95% CI]	24-h POD [95% CI]	24-h CSI
CAMS operational forecast	41.6 [34.6, 50.8]	0.92 [0.89, 0.95]	0.86
CAMS reanalysis (EAC4)	43.0 [33.5, 54.9]	0.72 [0.61, 0.82]	0.68
bias-corrected CAMS	30.3 [25.2, 36.6]	0.92 [0.89, 0.96]	0.88
persistence	13.0 [9.7, 17.2]	0.97 [0.96, 0.99]	0.95
FM (Chronos-Bolt)	14.7 [10.5, 20.6]	0.96 [0.93, 0.98]	0.94
FM (Chronos-2)	13.3 [10.1, 17.5]	0.97 [0.95, 0.99]	0.94

Table S15: **Cross-region robustness.** Benchmark MAE ($\mu\text{g}/\text{m}^3$) by country/subregion for raw CAMS, persistence and Chronos-2, with the Chronos-2 reduction over CAMS. The observation-driven nowcast beats CAMS in every region and for both pollutants. $\text{PM}_{2.5}$ regions are single stations (point estimates); the PM_{10} row is the 19-station Abu Dhabi network.

Region	Pollutant	Sites	raw CAMS	Persistence	Chronos-2	FM gain (%)
Bahrain (Manama)	PM2.5	1	25.9	19.7	17.0	34.4
Kuwait (Kuwait City)	PM2.5	1	47.0	22.8	22.0	53.2
Qatar (Doha)	PM2.5	1	21.7	15.9	13.7	36.9
Saudi Arabia, E. coast (Dhahran)	PM2.5	1	24.8	13.2	11.3	54.4
Saudi Arabia, Red Sea (Jeddah)	PM2.5	1	12.4	10.8	9.9	20.2
UAE, Abu Dhabi network	PM10	19	43.0	13.0	13.3	69.1

not the 19 stations or the ~ 80 distinct origin days (about 65 origins per station at stride 18 h, overlapping across stations) but the 12 multi-day dust *episodes* (maximal runs of origins separated by no more than 36 h). We therefore recomputed the headline PM_{10} comparison with an *episode-block bootstrap* that resamples whole episodes with replacement (2000 resamples, 12 episode blocks), which is the correct scale for synoptic dependence. Resampling at this coarser scale changes the per-model intervals (raw CAMS 43.0, 95% CI [38.1, 47.7]; Chronos-2 13.3, [8.7, 17.8]): because episode resampling holds the 19-station panel fixed, between-station variance, the dominant term in the station cluster bootstrap, is not resampled, and the raw-CAMS interval is accordingly narrower than its station-cluster counterpart rather than wider. The raw-CAMS–Chronos-2 pooled-MAE gap nonetheless remains $29.8 \mu\text{g}/\text{m}^3$ (95% CI 28.7–31.5) and excludes zero, because CAMS trails the nowcast by a similar margin in every episode (Table S16). We do not claim either one-way bootstrap captures the full station $\times$ episode dependence; the true joint interval is likely wider than either marginal. The effect size ($d_z = 1.49$) is nonetheless large enough that the sign is secure under episode-scale resampling. What no resampling can repair is external generalisation: the reported p -values speak to within-season, within-region skill only, and transfer to other regions or years remains to be tested.

Table S16: **Episode-block bootstrap** (PM_{10}). Pooled MAE ($\mu\text{g}/\text{m}^3$) and the raw-CAMS–Chronos-2 gap with 95% intervals from resampling whole synoptic dust episodes, which is the correct scale for synoptic temporal dependence. Episodes are defined operationally as maximal runs of forecast origins separated by no more than 36 h, which partitions the September–December 2023 record into 12 multi-day blocks; blocks, not origins, are the resampling units (2000 resamples). Because episode resampling holds the station panel fixed, between-station variance is not resampled and the raw-CAMS interval is narrower than its station-cluster counterpart; the paired gap nonetheless excludes zero because CAMS trails the nowcast in every one of the 12 episodes.

Quantity	Estimate	95% CI (episode-block)
raw CAMS MAE	43.0	[38.1, 47.7]
bias-corrected CAMS MAE	30.3	[23.7, 37.5]
Chronos-2 MAE	13.3	[8.7, 17.8]
Chronos-Bolt MAE	14.7	[9.7, 18.9]
persistence MAE	13.0	[8.2, 18.0]
seasonal-naive MAE	22.0	[12.9, 30.1]
raw CAMS – Chronos-2 gap	29.8	[28.7, 31.5]

S7 Grid representativeness versus systematic bias (PM₁₀)

Because the CAMS grid is coarse (0.75° , ≈ 80 km), point monitors within one cell share an identical CAMS value, so the spread of their observed means bounds the point-versus-cell representativeness error. Table S17 lists the six cells containing two or more PM₁₀ monitors: the within-cell station standard deviation has median $8.4 \mu\text{g m}^{-3}$ but reaches $\sim 40 \mu\text{g m}^{-3}$ in the two strong-gradient urban dust cells, so representativeness variability is concentration-dependent and material at the hotspots. We do not claim this spread is negligible; regulatory monitors are preferentially sited in urban/roadside dust hotspots, so part of the large single-station biases (e.g., $-121 \mu\text{g m}^{-3}$ at Al Mafrq) is genuine sub-grid sampling mismatch rather than an emission deficiency, and we therefore report the network-mean bias rather than the extreme. The key control against a representativeness artefact is not this bound but the direct test in Table S18: restricting to the nine best-represented stations (those with $|\text{CAMS bias}| \leq 15 \mu\text{g m}^{-3}$), where sampling mismatch is smallest, the observation-driven advantage is undiminished (Chronos-2 lowers MAE over CAMS by 68%, versus 69% over all 19). Because the headline holds where representativeness error is smallest, it is not an artefact of a few badly-sited monitors.

Table S17: **Representativeness bound (PM₁₀)**. For each 0.75° CAMS cell containing ≥ 2 monitors (identical CAMS value), the number of monitors, their mean observed concentration, and the within-cell range and standard deviation of station means (a direct estimate of point-versus-cell representativeness variability). Stations are assigned to the nearest 0.75° grid centre, matching the extraction in Section 2; the deposited helper script has been corrected to use the same rule. Sixteen of the 19 stations share a cell with at least one other, so their CAMS covariate series are identical, which is why a leave-one-grid-cell-out stage is included in the audit (Table S6, Table S24). The spread is concentration-dependent: the median within-cell standard deviation is $8.4 \mu\text{g m}^{-3}$ but reaches $37.5\text{--}43.0 \mu\text{g m}^{-3}$ in the two high-dust urban cells (within-cell ranges of 69 and $86 \mu\text{g m}^{-3}$), so representativeness is material at hotspots. The headline is therefore validated on the well-represented station subset (Table S18), not by a magnitude bound alone.

Cell lat	Cell lon	Monitors	Obs mean	Within-cell range	Within-cell SD
24.00	52.50	2	71	16	11.6
24.00	54.00	3	82	69	37.5
24.00	54.75	3	131	86	43.0
24.00	55.50	4	79	12	5.3
24.75	54.00	2	100	7	4.8
24.75	54.75	2	106	6	4.4

Table S18: **The headline survives on well-represented stations (PM₁₀)**. Benchmark MAE ($\mu\text{g/m}^3$) split by CAMS representativeness: the nine stations with small CAMS bias ($|\text{bias}| \leq 15 \mu\text{g m}^{-3}$, mean $|\text{bias}| 6.1 \mu\text{g m}^{-3}$; least affected by sub-grid siting) versus the ten high-bias stations. The Chronos-2 reduction over raw CAMS is essentially identical in both subsets (68% vs 70%) and matches the full-network value (69%), so the observation-driven advantage is not driven by point-versus-cell representativeness error at hotspot monitors.

Subset	Stations	raw CAMS	Persistence	Chronos-2	FM gain (%)
All	19	43.0	13.0	13.3	69
Low CAMS bias ($ \text{bias} \leq 15$)	9	26.7	8.5	8.6	68
High CAMS bias	10	57.2	16.8	17.3	70

S8 Independent-model cross-check: MERRA-2 (PM₁₀)

To test whether the observation-driven advantage is specific to the CAMS/ECMWF modelling system or to its 0.75° resolution, we repeated the PM₁₀ comparison against NASA MERRA-2 (M2T1NXAER hourly aerosol surface diagnostics, $0.5^\circ \times 0.625^\circ$; GEOS/GOCART, an independent model and finer grid than EAC4), retrieved from the GES DISC archive and sampled at the grid cell nearest each station and the nearest hour to each benchmark valid time (within 90 min). The M2T1NXAER surface collection resolves dust only as a total field (to $\sim 20 \mu\text{m}$ diameter) and a PM_{2.5}-mode field, with no direct sub- $10 \mu\text{m}$ (PM₁₀) dust diagnostic, so a single MERRA-2 PM₁₀ is ambiguous. We therefore evaluate two reconstruction scenarios. The total-dust scenario uses DUSMASS + SSSMASS + BCSMASS + 1.4 OCSMASS + 1.375 SO4SMASS ($\times 10^9$ to $\mu\text{g}/\text{m}^3$); because total dust includes coarse particles beyond $10 \mu\text{m}$ diameter, it over-predicts PM₁₀ (mean bias $+127 \mu\text{g m}^{-3}$). The PM_{2.5}-mode scenario replaces total dust and sea salt with their PM_{2.5}-mode diagnostics (DUSMASS25, SSSMASS25); by omitting the $2.5\text{--}10 \mu\text{m}$ coarse dust that belongs to PM₁₀, it under-predicts (mean bias $-35 \mu\text{g m}^{-3}$). For each scenario we also remove the per-station mean observation–MERRA offset (bias correction), which absorbs the reconstruction offset and any model mean bias. On the same 9,872 matched forecast points, bias-corrected MERRA-2 PM₁₀ MAE spans 25.9 (PM_{2.5}-mode scenario) to $101 \mu\text{g m}^{-3}$ (total-dust scenario). This is not a strict mathematical bracket on MAE, because bias-corrected MAE need not be monotonic in the reconstructed concentration. The relevant robustness point is instead conservative: even the most favourable reconstruction scenario remains about $1.95\times$ the observation-driven nowcast error (25.9 vs. $13.3 \mu\text{g m}^{-3}$) and only modestly better than bias-corrected CAMS ($30.3 \mu\text{g m}^{-3}$; Table S19). An independent, finer mechanistic reanalysis therefore does not close the gap under these coarse-dust treatments: a model that can even beat CAMS after bias correction still cannot track short-horizon point concentrations without the recent observation, because it lacks that information, not because of its grid or family.

Table S19: **Independent-model cross-check (PM₁₀, MERRA-2), two reconstruction scenarios.** MAE ($\mu\text{g}/\text{m}^3$) with station cluster-bootstrap 95% intervals and mean bias, on the same 9,872 matched PM₁₀ forecast points (48 of the 9,920 benchmark points lack a MERRA-2 value within the 90-min matching window). The PM_{2.5}-mode scenario under-predicts coarse dust, whereas the total-dust scenario over-predicts PM₁₀. The resulting MAE span is not a strict mathematical bound, but even the most favourable bias-corrected scenario is about $1.95\times$ the observation-driven nowcast and only modestly better than bias-corrected CAMS. *Provenance:* MERRA-2 M2T1NXAER granules are retrieved from the NASA GES DISC archive and are not redistributed in the deposited archive; reproducing this table requires a NASA Earthdata login and the collection and matching parameters documented in `data_retrieval/RETRIEVAL.md` of the archive; no retrieval client is shipped.

Model	MAE [95% CI]	Mean bias
MERRA-2 total-dust scenario, raw	136.1 [119.9, 152.5]	+127.2
MERRA-2 total-dust scenario, bias-corrected	101.3 [91.1, 112.4]	0.0
MERRA-2 PM _{2.5} -mode scenario, raw	43.2 [34.0, 55.0]	−34.7
MERRA-2 PM _{2.5} -mode scenario, bias-corrected	25.9 [22.4, 30.3]	0.0
CAMS reanalysis (reference)	43.0	−23.6
persistence (reference)	13.0	+2.0
Chronos-2 (reference)	13.3	+3.1

S9 Reproducibility details

Foundation models were run zero-shot from public checkpoints with the `chronos-forecasting` (v2.3.1) and `timesfm` packages on PyTorch 2.x (CPU): `amazon/chronos-2`, `amazon/chronos-bolt-small`, and `google/timesfm-2.5-200m` (six-site cross-check). Point forecasts are the predictive median (0.5 quantile); no probabilistic post-processing was applied. Inputs are the raw observed series in native $\mu\text{g}/\text{m}^3$ with the model’s internal scaling; no external normalisation was added. Context windows require $> 95\%$ non-missing coverage and are gap-filled by forward/backward fill before inference; origins whose targets contain any missing value are dropped. All stochastic steps use fixed, recorded seeds (0 unless stated): the gradient-boosting transfer (`HistGradientBoostingRegressor`, 300 iterations, learning rate 0.05, max depth 6) and every bootstrap within a given table-generation script; confidence intervals in different tables therefore come from independently seeded bootstrap draws over the same data, which accounts for sub-tenth differences between, e.g., Tables S2 and S14. Where a robustness table was generated as an independent matched-table run, the supplement preserves the table-specific seed record and output so that small bootstrap-CI differences can be reproduced. Rolling origins are equally spaced (up to 120 per station, stride 6 steps). The bias-corrected baseline removes the rolling mean of the past- K observation–CAMS residual; seasonal-naive uses the value 24 h earlier (dropped when missing); the CAMS-residual hybrid forecasts the residual series and adds it back to the aligned CAMS field at each target step. Chronos-2 and Chronos-Bolt are the released model sizes named above. The pretraining corpora of public foundation models are not fully auditable, so prior exposure to public air-quality time series, including OpenAQ-like data, cannot be completely excluded; this is why all model claims are interpreted against persistence, seasonal-naive and CTM-only references rather than as pure zero-shot generalisation. MERRA-2 M2T1NXAER (collection 5.12.4) hourly surface aerosol diagnostics were retrieved from NASA GES DISC and reconstructed to surface PM_{10} as described above; the nearest grid cell and nearest hour (within 90 min) to each benchmark valid time were used. Machine-readable processed tables, consistency checks and script entry points are in the public repository at github.com/bisu9082/serra_camscast.

Table S20: **Stability of the cold-start transfer under source-set resampling.** The estimates in Tables S6, S7 and S10 come from a deterministic fit (internal early stopping disabled, `random_state = 0`), which reproduces bit-for-bit under any seed; a seed-based stability check is therefore undefined for the estimator actually used. We instead resample the quantity that is actually stochastic, the set of source stations, drawing 100 bootstrap source sets per target and refitting (an earlier version used 12; the ratio below is smaller with the larger sample and we report the larger sample). The coordinate-bearing specification is 3.5 times the more variable at zero buffer and 4.6 times at 100 km, in the same direction as every other diagnostic in the audit: which stations happen to be available matters far more to the coordinate-bearing fit than to the coordinate-free one. Entries are the pooled gain over raw CAMS, mean $\pm$ standard deviation across bootstrap replicates, with the 2.5th–97.5th percentile interval.

R (km)	Feature set	Pooled gain (%)	SD (pp)	95% interval (%)	SD ratio
0	with lat/lon	+8.3	7.49	[-7.0, +20.0]	3.5×
0	coordinate-free	+18.9	2.13	[+15.4, +23.3]	
100	with lat/lon	-28.8	11.61	[-52.2, -8.2]	4.6×
100	coordinate-free	+6.3	2.52	[+1.3, +10.2]	

Table S21: **Feature ablation: where the coordinate-free gain comes from (PM₁₀, distance buffer only).** Removing latitude and longitude leaves the day-of-year sinusoids carrying the entire gain. Over a single-year record those sinusoids are a bijection onto the date, and at a 100 km buffer every target timestamp is still present at some surviving source, so the “coordinate-free” gain under a spatial buffer is in part a lookup of the network-wide residual at the target’s own timestamp. This is the motivation for the time-blocking control in Table S6. Intervals are station cluster bootstraps.

Feature set	$R = 0$		$R = 100$ km	
	Pooled (%) [95% CI]	Median (%), Win	Pooled (%) [95% CI]	Median (%), Win
CAMS only	+11.7 [-0.4, +21.6]	-0.4, 9/19	-1.2 [-16.0, +8.3]	-4.5, 9/19
+ hour	+12.1 [-0.6, +21.8]	-1.6, 9/19	-3.4 [-19.4, +7.4]	-6.6, 9/19
+ day-of-year	+22.3 [+10.9, +30.7]	+31.0, 16/19	+10.7 [-1.8, +19.4]	+13.0, 12/19
coordinate-free	+22.3 [+10.3, +31.1]	+30.7, 16/19	+9.3 [-3.8, +18.5]	+13.0, 12/19
+ lat/lon (full)	+12.0 [-9.8, +25.9]	+15.6, 16/19	-39.3 [-89.1, -5.6]	-0.6, 9/19

Table S22: **Paired comparison of the two feature sets (PM₁₀).** Per-station transfer MAE, coordinate-bearing minus coordinate-free, so a positive difference favours the coordinate-free specification. Under the full spatiotemporal control the two are indistinguishable at zero buffer and separate only marginally at the widest buffer: coordinates are not harmful in themselves, they are fragile, and the buffer is what exposes that. The distance-buffer-only design overstates the separation at every radius.

R (km)	Design	Mean diff. ($\mu\text{g m}^{-3}$)	95% CI	Coord.-free better	d_z	Wilcoxon p
0	buffer only	+4.66	[-2.55, +12.95]	8/19	+0.26	0.679
0	buffer + time	+0.10	[-2.94, +3.19]	9/19	+0.01	0.829
25	buffer only	+5.30	[-1.61, +11.69]	12/19	+0.34	0.145
25	buffer + time	+1.53	[-1.05, +3.92]	13/19	+0.26	0.210
50	buffer only	+9.94	[+2.28, +17.79]	13/19	+0.57	0.045
50	buffer + time	+4.81	[-0.29, +10.60]	13/19	+0.39	0.113
100	buffer only	+22.38	[+8.39, +36.59]	14/19	+0.68	0.018
100	buffer + time	+6.41	[-0.11, +13.51]	14/19	+0.41	0.066

Table S23: **Is the transfer gain predicted by the target’s own CAMS bias?** Spearman rank correlation between |CAMS bias| at the held-out station and its transfer gain, by buffer radius. At zero buffer there is no association, which was the basis for the earlier claim that target-site bias is not a usable screening variable. The rank correlation rises with radius and is largest at the radius that corresponds to an unmonitored site, but only the spatial-only $R = 100$ km cell survives Benjamini–Hochberg correction across the sixteen cells of this family ($q = 0.023$); the full-control cell that an operational rule would have to rest on does not ($q = 0.12$). We therefore report the association as a hypothesis the data are consistent with, not as an established second gate, and the rule defended in the main text is the distance gate alone. The coordinate-free specification is the less bias-sensitive of the two at wide buffers.

Design	Feature set	$R = 0$	$R = 25$	$R = 50$	$R = 100$
buffer only	with lat/lon	-0.13 (0.586)	-0.17 (0.495)	+0.26 (0.286)	+0.68 (0.001)
buffer only	coordinate-free	+0.03 (0.904)	+0.34 (0.156)	+0.34 (0.152)	+0.52 (0.023)
buffer + time	with lat/lon	+0.28 (0.251)	+0.32 (0.188)	+0.34 (0.158)	+0.53 (0.021)
buffer + time	coordinate-free	+0.39 (0.098)	+0.32 (0.178)	+0.23 (0.348)	+0.26 (0.286)

Table S24: **Two further controls** (PM₁₀). *Upper block*, leave-one-grid-cell-out: every source sharing the target’s 0.75° CAMS cell is withheld, removing the identical-covariate path created by grid sampling (16 of 19 stations share a cell with at least one other, at separations up to 46 km). The gain is undiminished, so grid-cell sharing inflates the apparent collapse of the distance-buffered ladder without generating the transfer signal. Table S33 shows the same conclusion holds under seven positions of the grid origin. *Lower blocks*, coordinate permutation placebo under the specification this paper reports (flat-line QC at $K = 6$, quantile time blocks, ten-day guard, full buffer + time + cell control), on the log error ratio Δ . Each station keeps a constant coordinate pair but the pairs are permuted among stations, so the coordinates are geographically wrong while every other feature is untouched. Ninety-nine permutations are run at each radius, so the smallest attainable one-sided p is 0.010; an earlier version used 30 draws, at which 0.032 was the floor and neither result could be resolved below 0.05. The sign of the true-minus-scrambled difference changes along the ladder, which is the reading the placebo is for. At $R = 0$ the true assignment sits at the 59th percentile of its own null and is indistinguishable from a wrong one (two-sided $p = 0.90$): with every non-cellmate neighbour available the coordinates add nothing the day-of-year term does not already supply, a conclusion the feature ablation of Table S21 reaches independently. At $R = 100$ km the true assignment sits at the 10th percentile and 89 of 99 geographically wrong assignments beat it, because the target lies outside the region its surviving sources span and the learner extrapolates along the coordinate axes; that direction is clear but not itself significant (one-sided $p = 0.11$ for “true worse”, two-sided $p = 0.21$). We report the sign change and not a significance claim. The earlier buffer-only placebo at $R = 0$ (200 draws) is not carried forward, because it was computed under the pipeline this paper rejects — a three-day guard, calendar blocks and no flat-line control — and mixing it with these would mix both the specification and the scale. The permutation null is summarised rather than bootstrapped, so no cluster-bootstrap interval is given for it (n/a); intervals elsewhere are station cluster bootstraps.

Control	Feature set / replicate	Pooled gain (%)	95% CI	Win	d_z
<i>leave-one-grid-cell-out (mean 16.4 surviving sources)</i>					
cell only	with lat/lon	+15.2	[-4.8, +26.8]	16/19	+0.39
cell only	coordinate-free	+18.1	[+5.0, +27.8]	16/19	+0.59
cell + time	with lat/lon	+16.8	[+10.3, +24.2]	18/19	+0.87
cell + time	coordinate-free	+14.1	[+5.2, +21.4]	16/19	+0.67
<i>coordinate permutation placebo, full control, 99 permutations per radius</i>					
		$R = 0$		$R = 100$ km	
true assignment	median Δ	+0.1316	(13/19)	+0.0338	(10/19)
permutation null	mean $\pm$ SD	+0.1243 $\pm$ 0.0561		+0.1097 $\pm$ 0.0604	
null 2.5–97.5%		[+0.0221, +0.2291]		[+0.0143, +0.2062]	
percentile of true		59th		10th	
scrambled $\geq$ true		41 of 99 (41%)		89 of 99 (90%)	
one-sided p , true worse		0.590		0.110	
two-sided p		0.900		0.210	
reading		coordinates inert		coordinates a liability	

S10 Detection skill: WHO 24-h mean and short-duration episodes

Error in $\mu\text{g m}^{-3}$ matters for warnings only insofar as it changes which exceedances are flagged, so we also evaluate threshold detection (Figure 4 of the main text), reported two ways to keep the health interpretation rigorous. First, as a *formal WHO 24-h mean* exceedance: each origin’s eight-step forecast mean (a prospective 24-h average) is compared against the observed 24-h mean. At the PM_{10} guideline ($45 \mu\text{g m}^{-3}$), exceedances are common in this late-2023 (September–December) dust-period subset (1119 of 1240 origin-level 24-h means, 90.2%), so an always-alarm rule would already reach CSI 0.90. Against that high-base-rate reference, raw CAMS detects about seven in ten 24-h exceedance windows (POD 0.72, 95% CI 0.61–0.83; CSI 0.68), whereas observation-driven nowcasts reach near-complete detection (foundation model POD 0.97, 0.95–0.99; persistence 0.97) at comparable false-alarm ratio (Table S12). Because the base rate is high, raw CSI flatters every method, so we also report chance-corrected scores computed from the same contingency tables: the equitable threat score is 0.06 for raw CAMS (essentially no skill beyond chance despite a CSI of 0.68) against 0.53–0.56 for the observation-driven nowcasts (persistence 0.56, seasonal-naïve 0.45, Chronos-Bolt 0.53, Chronos-2 0.54) and 0.27 for bias-corrected CAMS (Table S12). Chance correction therefore sharpens rather than softens the contrast. Second, as instantaneous three-hourly operational tiers, which are *not* WHO health metrics but capture short-duration episodes relevant to same-day advisories. On a severe short-duration dust tier ($\geq 150 \mu\text{g m}^{-3}$, 3-h), raw CAMS detects only 6% of episodes (POD 0.06, 0.03–0.12), reflecting its systematic dust under-prediction, while a recent-observation nowcast recovers roughly two-thirds (foundation model POD 0.67, 0.49–0.76; persistence 0.68); we phrase this as low retrospective detection skill for EAC4 in this benchmark rather than a general operational claim. This target-based detection analysis should not be read as a first-onset lead-time test: a persistence-like nowcast can score well once dust is already elevated while still missing a sudden onset from a clean recent history, the classic forecaster’s dilemma for observation-conditioned warnings. The three-hourly detection gap persists across the 24 h horizon (Figure 4 of the main text). The $\text{PM}_{2.5}$ guideline ($15 \mu\text{g m}^{-3}$) is exceeded almost ubiquitously (545 of 600 origin-level 24-h means, base rate 90.8%, at which an always-alarm rule already scores CSI 0.91), so 24-h detection is uniformly high, including raw CAMS, and does not discriminate: raw CAMS and persistence (CSI 0.90 and 0.89) do not beat that reference at all, and the methods that do clear it, seasonal-naïve (CSI 0.93) and the two foundation models (CSI 0.93), exceed it only marginally, with the seasonal-naïve reference matching them (chance-corrected ETS 0.44 for seasonal-naïve and 0.42–0.45 for the foundation models, against 0.20 for raw CAMS, Table S12); and no method reliably flags rare, sharp severe- $\text{PM}_{2.5}$ spikes ($\geq 75 \mu\text{g m}^{-3}$, 3-h; CSI ≤ 0.21 for all), a limit of short-horizon autoregressive prediction for rare sharp $\text{PM}_{2.5}$ spikes, whose composition (fine-mode dust, secondary or combustion aerosol) is not resolved by our data (raw CAMS reaches a higher severe- $\text{PM}_{2.5}$ POD only by over-warning, FAR 0.80, so its CSI is no better). Throughout, persistence matches the foundation model, reinforcing that the lever is the observation stream, not the model.

S11 Robustness: operational forecast and cross-region consistency

Two checks address the largest external-validity concerns for a reanalysis-based, single-region benchmark. First, to test whether using EAC4 rather than a real-time product drives the result, we repeated the PM_{10} benchmark with the archived CAMS operational forecast (initialised 00/12 UTC), sampled at each origin from the most recent cycle at or before the origin and evaluated on the *same* rolling origins as the reanalysis (Table S14, Figure S1). The operational forecast is statistically in-

distinguishable from the reanalysis (MAE $41.6 \mu\text{g m}^{-3}$, 95% CI 34.6–50.8, vs. 43.0, 33.5–54.9) and, like it, roughly three times the observation-driven nowcast error (13.0–13.3); both CAMS products stay flat near $40\text{--}45 \mu\text{g m}^{-3}$ across the 24-h horizon while the observation-driven error rises from ~ 4 to $\sim 21 \mu\text{g m}^{-3}$ with lead (Figure S1b). On WHO 24-h mean detection the operational forecast flags more exceedances than the reanalysis (POD 0.92 vs. 0.72) but only by over-forecasting dust, so its balanced skill remains below the observation-driven nowcasts (CSI 0.86 vs. 0.94–0.95). The choice of CAMS product therefore does not change the regime map: the two CAMS products are indistinguishable on error, and although the operational forecast scores higher on 24-h detection than the reanalysis (CSI 0.86 vs. 0.68), it does so by over-forecasting and still trails every observation-driven nowcast. The result is likewise not specific to the CAMS model family: for an independent NASA MERRA-2 reanalysis (finer resolution), bias-corrected PM_{10} reconstruction scenarios span $26\text{--}101 \mu\text{g m}^{-3}$ across coarse-dust treatments. The most favourable scenario outperforms raw CAMS and is modestly better than bias-corrected CAMS (25.9 vs. $30.3 \mu\text{g m}^{-3}$), but it remains roughly twice the observation-driven nowcast ($13 \mu\text{g m}^{-3}$; Methods; SI, Table S19). Second, the observation-driven advantage is present in every national network sampled: for $\text{PM}_{2.5}$, Chronos-2 lowers MAE over CAMS at each of the four countries and both coasts, by 20% (Jeddah, Red Sea) to 54% (Dhahran) (Table S15). The headline is thus not an artefact of the reanalysis product or of a single national network, though a fully external region and season outside the Arabian Peninsula remains the strongest outstanding test.

S12 Atmospheric interpretation: why the observation stream dominates

The pattern of wins and losses is physically coherent rather than a generic machine-learning artefact, and it says something about the pollution problem itself. Gulf PM_{10} is dominated by episodic, coarse-mode mineral dust whose emission is strongly influenced by surface wind gusts (including Shamal events (Yu et al., 2016)), soil state and land use at scales far below an 0.75° grid cell (Draxler et al., 2001; Gandham and Hoteit, 2020), so a global CTM inherits a structural under-prediction of coarse dust at monitors (network-mean bias $-23.6 \mu\text{g m}^{-3}$ here, with the largest deficits at urban and roadside sites) that assimilation of column aerosol optical depth does not, by itself, constrain at the surface (Inness et al., 2019; Rémy et al., 2019); part of this deficit is also a documented property of the EAC4 PM_{10} diagnostic, whose dust representation truncates the coarsest size modes, and regional evaluations over the Middle East report a similar surface shortfall for the global assimilation products (Rémy et al., 2019; Ukhov et al., 2020). $\text{PM}_{2.5}$, by contrast, is over-predicted (all five sites positive, up to $+36.9 \mu\text{g m}^{-3}$), a bias to which several mechanisms can plausibly contribute, among them anthropogenic-emission uncertainty and over-represented sea salt at these coastal sites, though our data do not resolve their relative shares. Aerosol water is not among them: the IFS-AER particulate diagnostic is defined on a dry basis, with sea-salt mass converted from its 80% relative-humidity reference before summation (Rémy et al., 2019). Whatever the mechanism split, the resulting sign reversal between pollutants defeats any single blanket correction. At the same time, dust loading during an event is strongly autocorrelated over hours: once a plume is over a station, its own recent record is the best available predictor of the next 3–24 h, which is why persistence captures most of the observation-driven skill and why the advantage decays with lead time as that autocorrelation fades. The regime map is therefore an atmospheric statement as much as a methodological one: the CTM field retains information where local memory is weakest, at abrupt $\text{PM}_{2.5}$ extremes arriving by transport, at unmonitored locations and (in principle) at episode onset. It is redundant where the coarse-dust signal the model cannot

resolve is already written into the local observation stream.

S13 Learner sensitivity: is the verdict a property of one model?

The audit above uses a single histogram gradient-boosting regressor. A reviewer of an earlier version asked whether its verdict is a property of that learner rather than of the design, so we repeated the full-control ladder with four learners spanning a wide capacity range: the reported gradient boosting at depth 6, the same at depth 2, a 200-tree random forest, and ridge regression on standardised features with a cross-validated penalty. Every learner is deterministic (early stopping off, fixed `random_state`, and `n_jobs` = 1 for the forest so that the result does not depend on thread count), and the depth-6 rows reproduce the values in Table S6 to the last reported digit, which is also our execution check on the reimplement.

Two questions have to be kept apart here, and an earlier version of this supplement ran them together. The first is whether each learner, on its own, reproduces a positive transfer verdict. The second is whether the learners differ from one another. Answering the second by inspecting whether each learner’s separate interval excludes zero is the error this manuscript objects to elsewhere (Gelman and Stern, 2006), and we had committed it here. The four learners are fitted on the same 19 stations, so the difference between two of them is directly testable, paired by station.

Verdict reproducibility. At a 100 km buffer the *coordinate-free* median gain is positive with a bootstrap interval excluding zero under all four learners (+11.4, +14.5, +12.2, +17.9%), and the coordinate-bearing pooled gain is negative under all four (−3.6, −9.1, −21.5, −11.9%) with a worst station between −104 and −247% against −54 to −77% coordinate-free. The coordinate-bearing *median* gain stays positive under all four but its interval excludes zero only for the reported depth-6 model (+7.9% [+2.4, +12.2]); for the other three it spans zero. The coordinate-free positive verdict is therefore reproduced by every learner tested, and the coordinate-bearing one is not reproduced outside the reported model. That is the statement a practitioner inherits on choosing a learner, and it is the statement the manuscript makes.

Whether the learners themselves differ. Tested properly — paired Wilcoxon on the per-station gains against the depth-6 model, Benjamini–Hochberg across the three comparisons — the answer inverts the impression the intervals give. For the coordinate-bearing specification no learner differs from the reported one (median paired differences +2.9, −0.9 and −5.5 percentage points; p = 0.13, 0.54, 0.89; every q > 0.40). We therefore do *not* claim that the coordinate-bearing central estimate depends on the learner; what we claim, and all we claim, is that its verdict against zero is not reproduced. For the coordinate-free specification one difference is detectable: ridge returns a larger gain than the depth-6 tree, a median paired difference of −4.6 percentage points with 17 of 19 stations higher under ridge (p = 7×10^{-5} , q = 0.0002), while the depth-2 tree and the random forest do not differ (q = 0.12 and 0.33). Its direction reinforces the reported verdict rather than threatening it. Ridge is instructive for a separate reason: it is the one learner that cannot memorise a station identity in the tree sense, and it shows the same coordinate-bearing collapse, which locates the mechanism in extrapolation beyond the training coordinate hull rather than in tree capacity. The paired comparisons are in `empirical_learner_paired.csv`, regenerated by `code/empirical_learner_paired.py`.

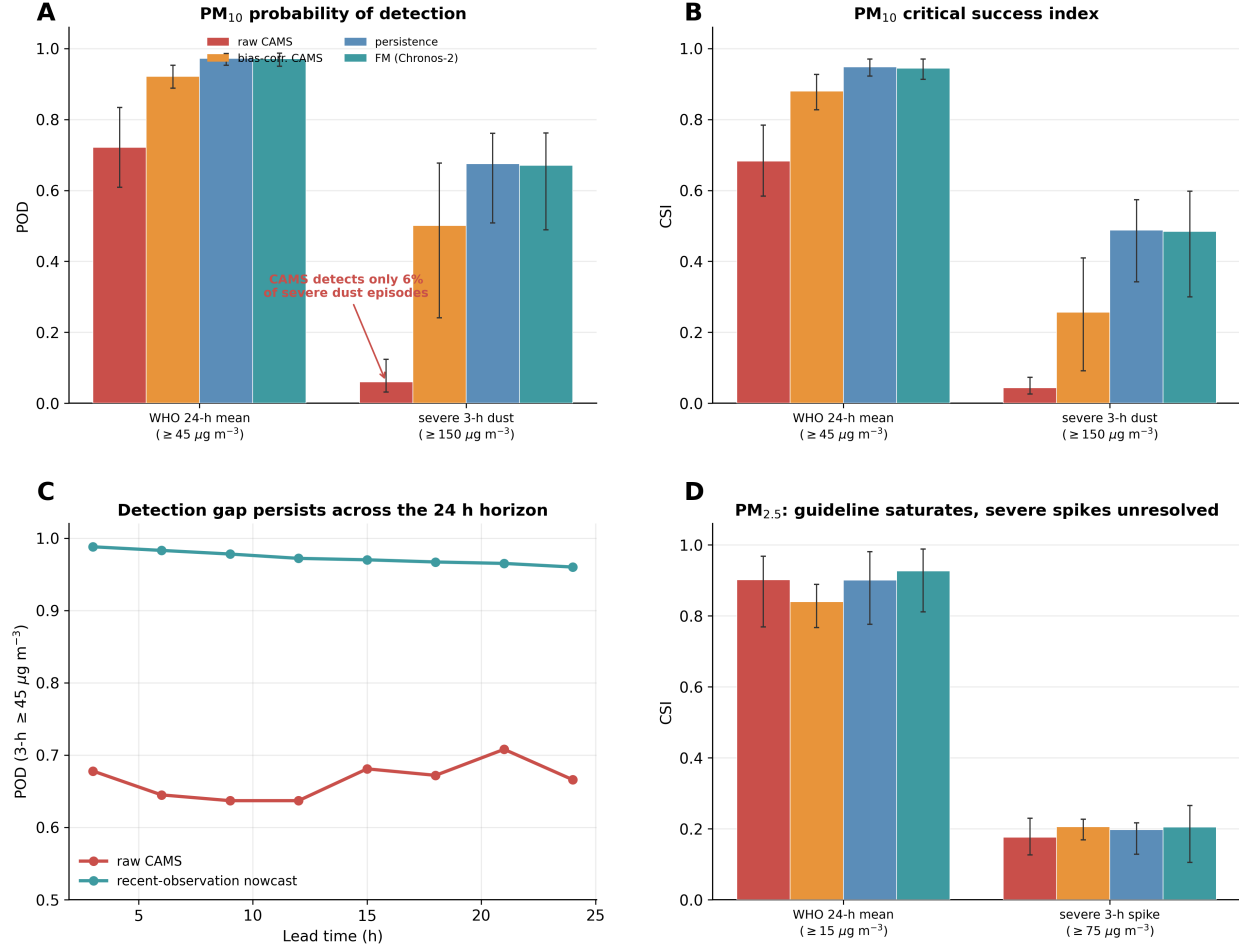


Figure S2: **Detection, separated by threshold type.** This figure supports the benchmark arm and is placed here so the main text can carry the audit. (a) PM₁₀ probability of detection (POD) for a *formal WHO 24-h mean* guideline exceedance ($\geq 45 \mu\text{g m}^{-3}$) and for a severe short-duration 3-h dust tier ($\geq 150 \mu\text{g m}^{-3}$; an operational, not health, threshold): the CAMS reanalysis detects about seven in ten 24-h guideline days and only 6% of severe 3-h dust episodes, whereas observation-driven nowcasts detect nearly all guideline days and about two-thirds of severe episodes. (b) Corresponding critical success index (CSI), with station cluster-bootstrap 95% intervals. (c) Three-hourly detection POD versus lead time. (d) PM_{2.5}: the 24-h guideline is exceeded almost ubiquitously and saturates detection, and no method reliably flags rare severe 3-h PM_{2.5} spikes. Persistence matches the foundation model throughout.

Table S25: **Learner sensitivity of the audit (full control: distance buffer + time blocks + guard band)**. Pooled and median per-station gain over raw CAMS, with the station cluster-bootstrap 95% interval on the median (2000 resamples), the win count, and the single worst station. “Reported” marks the learner used everywhere else in the manuscript; its four rows reproduce Table S6 to the last reported digit.

Learner	Pooled (%)	Median (%)	Median 95% CI	Win	Worst station (%)
<i>buffer R = 0 km, with lat/lon</i>					
Gradient boosting, depth 6 (reported)	+16.5	+14.4	[+6.7, +26.5]	17/19	-20.0
Gradient boosting, depth 2	+20.2	+23.2	[+14.6, +31.6]	16/19	-30.5
Random forest, 200 trees	+16.1	+17.8	[+9.3, +23.3]	18/19	-22.0
Ridge regression (standardised, CV-selected α)	+20.8	+22.2	[+12.1, +30.7]	16/19	-41.4
<i>buffer R = 0 km, coordinate-free</i>					
Gradient boosting, depth 6 (reported)	+16.8	+16.7	[+7.2, +24.8]	16/19	-49.1
Gradient boosting, depth 2	+20.5	+21.9	[+17.8, +26.5]	16/19	-45.9
Random forest, 200 trees	+15.4	+15.2	[+3.3, +23.7]	16/19	-44.2
Ridge regression (standardised, CV-selected α)	+21.5	+22.6	[+17.3, +32.5]	16/19	-33.7
<i>buffer R = 100 km, with lat/lon</i>					
Gradient boosting, depth 6 (reported)	-3.6	+7.9	[+2.4, +12.2]	15/19	-125.2
Gradient boosting, depth 2	-9.1	+2.1	[-31.2, +11.3]	10/19	-103.8
Random forest, 200 trees	-21.5	+6.4	[-52.5, +16.1]	12/19	-247.3
Ridge regression (standardised, CV-selected α)	-11.9	+6.4	[-73.2, +20.1]	10/19	-135.3
<i>buffer R = 100 km, coordinate-free</i>					
Gradient boosting, depth 6 (reported)	+10.2	+11.4	[+7.0, +20.3]	15/19	-77.2
Gradient boosting, depth 2	+12.0	+14.5	[+6.9, +19.1]	16/19	-67.0
Random forest, 200 trees	+10.7	+12.2	[+6.6, +19.0]	16/19	-68.4
Ridge regression (standardised, CV-selected α)	+15.3	+17.9	[+13.6, +23.4]	16/19	-54.1

S14 Does the grid-cell stage depend on where the CAMS grid origin sits?

The third control withholds every source station sharing the target’s 0.75° CAMS cell, because such stations carry an identical covariate series. Which stations share a cell is a property of the grid, and the grid’s origin belongs to the CAMS product rather than to the atmosphere. A reviewer asked whether the control is doing something real or is an artefact of where the cell boundaries happen to fall relative to this cluster of stations, so we shifted the grid origin over a lattice of offsets and repeated the audit at the two radii that bracket the argument.

Table S33 gives the result. Shifting the origin changes the partition substantially — the number of occupied cells moves between six and nine, and the number of station pairs sharing a cell between 15 and 31, so at the least favourable offset twice as many pairs are treated as covariate-sharing as at the grid we use. The verdict does not move with it. At $R = 0$ the median Δ spans +0.073 to +0.132 across the seven offsets, a range of 0.058 log units, which is a fifth of the width of the block-bootstrap interval on any one of them, and the win count stays between 10 and 13 of 19. At $R = 100$ km every offset returns the identical median +0.034 and the identical 10 of 19: once the buffer exceeds 100 km no surviving source shares a cell with its target under any origin, so the cell stage is inert there by construction. That is the quantitative form of the statement in the main text that the grid-cell stage binds only at narrow radii, and it follows from the cell size and the buffer without any fitting.

The stage is therefore not an artefact of the grid, and neither is it doing the work that carries the paper’s conclusion: no offset produces a significant transfer gain at either radius. What it removes

is a specific and identifiable confound at narrow buffers, and it should be reported for the same reason the buffer is.

Table S26: **Grid-origin sensitivity of the leave-one-grid-cell-out stage.** The origin of the 0.75° grid is shifted by the stated fraction of a cell in latitude and longitude; (0,0) is the CAMS grid the manuscript uses. “Pairs” counts station pairs assigned to the same cell, out of 171. Δ is the median log error ratio for the coordinate-bearing specification under the full control with a ten-day guard and quantile blocks.

Origin lat	offset lon	Cells	Pairs	$R = 0$ median Δ	wins	$R = 100$ km median Δ (wins)
0	0	9	15	+0.132	13/19	+0.034 (10/19)
0	-0.25	8	19	+0.092	12/19	+0.034 (10/19)
-0.25	-0.25	6	27	+0.073	10/19	+0.034 (10/19)
-0.25	-0.50	7	31	+0.073	12/19	+0.034 (10/19)
-0.50	0	8	24	+0.098	12/19	+0.034 (10/19)
-0.50	-0.25	8	24	+0.086	11/19	+0.034 (10/19)
-0.50	-0.50	9	28	+0.106	13/19	+0.034 (10/19)
<i>range across offsets</i>				0.058	10-13	0.000 (10-10)

S15 Relation to nearest-neighbour distance matching and to the area of applicability

Two instruments from this literature were suggested to us as alternatives to a buffer ladder: nearest-neighbour distance matching cross-validation (NNDM and its k -means variant kNNDM) and the area of applicability (AOA) with its dissimilarity index (DI). We computed both on these data. They do not replace the ladder; they explain what the ladder is doing and they corroborate its verdict from a different direction.

NNDM asks that the distance from a cross-validation test point to its nearest training point match the distance from a prediction location to its nearest training point, so that the validation answers the question the deployment poses. Table S26 gives both distributions for our design. At zero buffer the two coincide (median 6.5 km each, Wasserstein-1 distance 0.9 km): plain leave-one-station-out over this network validates *interpolation between neighbours 6.5 km apart*, which is why an unbuffered cold-start number cannot speak to an unmonitored site. The buffer moves the test-to-train distance to 43, 74 and 108 km while the training set’s own internal spacing stays near 7 km. A kNNDM fold assignment would select one point on that curve, the one matching a nominated prediction geometry; the buffer ladder reports the whole curve, and a deployment rule expressed in kilometres requires the whole curve, since “how far is my unmonitored site from the nearest monitor” is a design variable and not a fixed property of the network.

That said, the prediction geometry can be measured rather than nominated, and doing so locates the ladder against the deployment this paper is about. Drawing 20,000 locations uniformly from the network’s bounding box and taking each one’s distance to the nearest monitor gives a median of 35.0 km (interquartile range 22.0 to 50.0, maximum 101.2): that is what an unmonitored site in this region is actually like. Against it, plain leave-one-station-out at 6.5 km is optimistic by 28.5 km in the median. Every design we audit is at least as demanding as deployment: with the grid-cell stage in place, the median distance from a target to its nearest surviving source is 48.3 km at zero buffer, 62.7 km at 25 km, 73.7 km at 50 km, 84.0 km at 75 km and 108.1 km at 100 km. The

deployment geometry therefore falls below the narrowest rung of the ladder, so the ladder brackets it rather than approximating it, and the fold assignment a kNNDM search would return lies inside the range we already report. We ran the distance diagnostic and not the search because the search selects one rung and we need the curve; the assignment space is not the obstacle (with 19 stations and six folds it holds 6.9×10^{11} partitions).

The AOA asks a complementary question in feature space rather than in geographic space: whether a prediction location is similar enough to the training data for the model’s cross-validation error to apply to it. We computed the DI as the minimum standardised Euclidean feature distance from each target point to the source-station training set, normalised by the mean nearest-neighbour distance within that training set, with the threshold set at the 0.75 quantile plus 1.5 interquartile ranges of the training DI. (We use unweighted standardised features rather than importance-weighted ones. For the coordinate-bearing specification this is the conservative direction and the conclusion is safe under either choice: weighting by permutation importance can only raise the weight on features the model relies on, and the coordinate-bearing targets are already wholly outside the area of applicability unweighted. For the coordinate-free specification the direction is not determined, since weighting merely redistributes among the remaining features, and we do not claim one.) The result, Table S27, is a direct and independent statement of the reliability finding: with coordinates, every one of the 19 targets at a 100 km buffer lies wholly outside the area of applicability (median DI 6.8), whereas the coordinate-free specification keeps 89.5% of target points inside it (median DI 1.2) and no station predominantly outside. An AOA analysis would therefore have refused to issue the coordinate-bearing buffered prediction at all, which agrees with our conclusion that the coordinate-bearing correction is not the one to deploy, and reaches it without using any outcome data.

Table S27: **Nearest-neighbour distance structure of the buffer ladder.** For each buffer radius: the distance from each target station to its nearest surviving source station (the sample-to-prediction distance that a cold-start deployment faces), the nearest-neighbour distance *within* the surviving source set, and the Wasserstein-1 distance between the two distributions, which is the quantity kNNDM minimises. At $R = 0$ the two distributions coincide, so unbuffered leave-one-station-out validates interpolation.

R (km)	Target-to-source NN, median (km)	IQR	Within-source NN, median (km)	W_1 (km)
0	6.5	[5.9, 28.9]	6.5	0.9
25	43.3	[32.1, 56.6]	7.0	26.3
50	73.7	[69.8, 77.0]	7.0	51.6
100	108.1	[104.6, 109.0]	7.0	86.6

S16 Number-to-source audit

An earlier version of this manuscript quoted an equitable threat score of 0.72–0.75 for the observation-driven nowcasts where the computed value is 0.53–0.56; One of those two figures is the Heidke skill score of the same contingency tables (0.717 for persistence); the other corresponds to no column of any table we computed. We found it by checking the reported numbers against the artefacts by hand, and we have since automated as much of that check as can be automated. The script `number_audit.py` indexes every numeric value in the computed artefacts (all result JSON files, the aligned station records and the per-point tables, each also carried across the percentage/log-ratio transform), extracts every numeric literal in the manuscript, the supplement and the cover letter

Table S28: **Area of applicability of the cold-start correction.** Dissimilarity index (DI) of each target station’s prediction points relative to its source-station training set, and the share falling outside the area of applicability (DI above the training threshold). “Stations mostly outside” counts targets for which more than half the prediction points fall outside. With coordinates at a 100 km buffer the correction is inadmissible everywhere by this criterion; the coordinate-free specification is admissible almost everywhere.

Feature set	R (km)	Median DI	Points outside AOA (%)	Stations mostly outside
with lat/lon	0	0.83	33.7	6/19
with lat/lon	100	6.76	100.0	19/19
coordinate-free	0	0.50	4.4	0/19
coordinate-free	100	1.23	10.5	0/19

together with the sentence containing it, and reports the ones no artefact holds. Roughly 95% of the empirical literals match an artefact automatically. The remaining few dozen are quantities defined in text rather than computed — station identifiers, publication years, page numbers, sentinel codes, fixed hyperparameters, values quoted from the cited literature rather than produced here, and sample counts belonging to arms whose intermediate artefacts are not in the audited directory — and they are listed with their context in `number_audit_unmatched.csv` for manual review, which we have carried out. We give the proportion rather than the counts deliberately: the audit runs over the manuscript itself, so any edit changes the totals, and a count printed here would be stale the moment the sentence containing it was written. The exact figures for a given build are in the two CSV files, which are regenerated with the manuscript.

We state the limitation plainly, because it bounds what the audit can conclude. A value-presence check cannot catch a number quoted from the wrong column: 0.72 is a real Heidke skill score in our own tables, so an automated index would have accepted it. What catches that class of error is the context column, read by a person against the table the sentence cites. The audit narrows the reading list by about six sevenths; it does not replace the reading. Both files are in the repository.

S17 Which specification survives, and why the three inference units agree once the comparison is paired

At a 100 km buffer under the full three-fold control, neither feature set establishes a transfer gain on its own. The coordinate-bearing specification has a median $\Delta = +0.034$ across 19 stations, improves 10 of them, and clears neither test after correction (sign $q = 1.00$, Wilcoxon $q = 0.41$); the coordinate-free specification has $\Delta = +0.087$, improves 11, and likewise clears neither ($q = 0.78$ and $q = 0.77$). The block-bootstrap interval over the nine covariate cells spans zero for both ($[-0.618, +0.191]$ and $[-0.162, +0.244]$). Table S28 sets the two side by side under each of the three inference units that are defensible here.

An earlier version of this analysis went on to select between the two specifications by asking which of them cleared a threshold against zero under each unit, and reported that the three units disagreed. That comparison is not valid: the difference between significant and not significant is not itself significant (Gelman and Stern, 2006). What the question requires is the paired difference $\Delta_{\text{coordinate-free}} - \Delta_{\text{coordinate-bearing}}$, taken station by station over the same folds. Under that test the three units do not disagree. The coordinate-free specification is ahead at 13 of the 19 stations (sign test $p = 0.167$, Wilcoxon $p = 0.036$, $d_z = 0.46$), ahead in 7 of the 9 cells ($p = 0.180$), and the block-bootstrap interval on the paired difference is $[-0.000, +0.601]$ against $[-0.000, +0.567]$ for the

station bootstrap. All three point the same way and none of them separates the two specifications at zero buffer (11 of 19, $p = 0.648$). The apparent disagreement was an artefact of the comparison, not a property of the units.

What the units do change is how much weight a spatially coherent failure carries, and that remains worth reporting. Table S29 resolves Δ by cell. Under the coordinate-bearing specification the four stations of the eastern-edge cell 24.00°N/55.50°E fail together and fail hard (−1.16, −1.14, −1.14, −0.43; cell median −1.14), which is what extrapolation beyond the region the surviving sources span should look like. The same four stations under the coordinate-free specification give −0.22, −0.16, −0.08 and +0.14, a cell median of −0.12. Failures are not confined to that cell in either specification — 5 of 9 cells have a negative median with coordinates and 4 of 9 without — but the four largest failures in the whole design are the coordinate-bearing ones in that single cell. A station-level error bar counts that coherent failure four times, a cell median counts it once, and a cell block bootstrap counts it between zero and four times depending on the draw. With 19 stations in 9 cells no one of the three resolves this, and we do not claim that one does; we report all three because they are cheap to report together and because a single one of them, quoted alone, is how an audit certifies a transfer result it should not.

Table S29: **The two feature sets under three inference units (100 km buffer, full three-fold control), and the paired comparison between them.** The upper block gives each specification against zero, which is what an earlier version compared; the lower block gives the paired difference, which is the comparison the question actually requires. q values are Benjamini–Hochberg corrected across the 24 buffer $\times$ feature-set $\times$ design cells of Table S6, with the sign test and the Wilcoxon test corrected as separate families; The paired block gives both the uncorrected p and the Benjamini–Hochberg q across the four buffer radii at which the paired contrast is run, so that it receives the same treatment as the tests against zero; no cell of the paired family reaches $q < 0.05$.

Quantity	With lat/lon	Coordinate-free	Separates them?
<i>(i) each specification against zero, stations as 19 units</i>			
median Δ	+0.034	+0.087	no
stations improving	10/19	11/19	no
sign test q	1.00	0.78	neither passes
Wilcoxon q	0.41	0.77	neither passes
<i>(ii) each specification against zero, nine cell medians as units</i>			
median of cell medians	−0.059	+0.154	no
cells improving	4/9	5/9	no
<i>(iii) each specification against zero, block bootstrap over nine cells</i>			
median Δ , 95% CI	[−0.618, +0.191]	[−0.162, +0.244]	neither excludes zero
<i>(iv) paired difference, coordinate-free minus coordinate-bearing</i>			
median paired difference	+0.030		—
stations favouring coordinate-free	13/19 ($p = 0.167$, $q = 0.223$)		—
Wilcoxon p , q , d_z	0.036, 0.144, 0.46		—
cells favouring coordinate-free	7/9 ($p = 0.180$)		—
station bootstrap 95% CI	[−0.000, +0.567]		—
cell block bootstrap 95% CI	[−0.000, +0.601]		—

Table S30: **The paired comparison across the buffer ladder, corrected as a family.** The paired contrast $\Delta_{\text{coordinate-free}} - \Delta_{\text{coordinate-bearing}}$ is run at each of the four radii, so it is a family of four and is corrected as one. The direction is consistent from 25 km outward — 13 to 15 of 19 stations and 7 of 9 cells favour the coordinate-free specification at every radius beyond zero — but no cell reaches $q < 0.05$ under either test. We report the pattern and not a difference.

Radius	Stations favouring coordinate-free	Cells favouring coordinate-free	Sign p (q)	Wilcoxon p (q)
0 km	11/19	5/9	0.648 (0.648)	0.738 (0.738)
25 km	15/19	7/9	0.019 (0.077)	0.113 (0.151)
50 km	14/19	7/9	0.064 (0.127)	0.096 (0.151)
100 km	13/19	7/9	0.167 (0.223)	0.036 (0.144)

Table S31: **Cell-resolved transfer gain at a 100 km buffer under the full control.** Median Δ of the stations in each 0.75° CAMS cell, ordered by the coordinate-bearing value. Negative medians occur in five of the nine cells with coordinates and four without, but the four largest failures in the design are the coordinate-bearing ones in the eastern-edge cell, whose four stations fail together. This is the structure a station-level error bar cannot represent.

Cell	Stations	With lat/lon	Coordinate-free
24.00N/55.50E	4	−1.140	−0.120
23.25N/53.25E	1	−0.805	−0.204
24.00N/52.50E	2	−0.439	−0.269
24.75N/55.50E	1	−0.293	+0.699
23.25N/55.50E	1	−0.059	+0.154
24.00N/54.00E	3	+0.079	−0.531
24.00N/54.75E	3	+0.154	+0.154
24.75N/54.00E	2	+0.209	+0.259
24.75N/54.75E	2	+0.231	+0.239

S18 Three conventions inside the audit: the guard band, the block boundaries, and the flat-line control

Three choices inside the temporal control were made by convention in an earlier version of this analysis. Each is reported here with the measurement that should have set it.

The guard band. Three days was chosen to exceed the duration of a synoptic dust episode. That is a statement about the physics, not about the residual field the audit actually blocks. Table S30 gives the median across stations of the lag autocorrelation of the observation-minus-CAMS residual. It is +0.87 at three hours and falls below $1/e$ within a day, but it does not continue to decay: it plateaus near +0.20 from two days out to a week and reaches +0.08 only at ten days. The plateau is a seasonal and synoptic background rather than episode persistence, and it is what a guard band has to clear. We set the guard at ten days and report three days alongside it in Table S6.

The block boundaries. Calendar-equal blocks are only equal in time. This record has no observations at all in four months of 2023, so calendar blocks leave three of the six folds nearly empty at every station: at station 369959, for instance, they held 218, 9, 4, 7, 335 and 489 points. Across the 19 stations the smallest calendar fold holds 4 points and the largest 517, and the median within-station ratio of largest to smallest fold is 124 (range 113 to 129) — a nominal six-fold control that is three-fold in substance. Blocks taken at quantiles of each target’s own timestamps hold 170 to 190 points across all stations and folds, a within-station imbalance ratio of 1.01.

The flat-line control. Station 369963 reports exactly $80 \mu\text{g m}^{-3}$ for 51 consecutive three-hourly steps, and 15.5% of all observations repeat the previous value. Two mechanisms produce repeats and they are separable by concentration (Table S31). The record is integer-valued with a median absolute step of $3 \mu\text{g m}^{-3}$, so exact ties are expected wherever concentrations are low: ties of that kind should sit below the network median of $74 \mu\text{g m}^{-3}$ and be depleted in values above $100 \mu\text{g m}^{-3}$, and they are. Runs of two to ten steps have median values of 51 to 65 and only 9 to 20% of them exceed $100 \mu\text{g m}^{-3}$, against 29% for the record as a whole. A latched sensor has no such signature and can run far longer than any calm spell, and the two longest runs (51 and 22 steps, 153 and 66 hours) look like that. We therefore remove runs of six or more identical consecutive values, which is 1.2% of the record across 12 stations, and report a threshold of three as sensitivity.

The sensitivity matters, and it is the third of the three conventions this paper is about. A threshold of three deletes 9.9% of the record, and what it deletes is not a random sample: the removed observations have a median of $60 \mu\text{g m}^{-3}$ against the network’s 74, and only 9.8% of them exceed $100 \mu\text{g m}^{-3}$ against 28.5% of the record. Removing them raises the retained mean absolute CAMS error from 46.6 to $48.4 \mu\text{g m}^{-3}$ and the retained median concentration from 74.5 to 77.0. A residual correction has more room to help in a higher-error regime, and under the aggressive threshold the coordinate-free specification becomes significant again at zero buffer ($q = 0.011$) and at 100 km ($q = 0.035$), where under the conservative threshold nothing clears correction anywhere ($q \geq 0.251$). The paired comparison moves in the same way, from 13 of 19 stations favouring the coordinate-free specification at 100 km under the conservative rule (Wilcoxon $p = 0.036$) to 10 of 19 under the aggressive one ($p = 0.123$). We read this as the threshold creating a gain rather than revealing one, and we report both so a reader can disagree.

Table S32: **Residual autocorrelation by lag, which sets the guard band.** Median across the 19 stations of the lag autocorrelation of the observation-minus-CAMS residual, on the three-hourly grid with gaps reindexed. The decay is not monotone: a synoptic plateau near +0.2 persists from two days to a week before the correlation reaches +0.08 at ten days.

Lag (h)	Lag (days)	Median autocorrelation
3	0.12	+0.870
6	0.25	+0.715
12	0.50	+0.497
24	1.00	+0.314
48	2.00	+0.177
72	3.00	+0.195
96	4.00	+0.224
120	5.00	+0.188
168	7.00	+0.241
240	10.00	+0.081

Table S33: **Repeated values separate by concentration.** For each run length of identical consecutive observations: the number of such runs, the median concentration at which they occur, and the share exceeding $100 \mu\text{g m}^{-3}$. Runs up to ten steps sit below the network median and are depleted in high values, which is the signature of integer reporting at a median step of $3 \mu\text{g m}^{-3}$; a latched instrument would carry no such signature. The two runs of eleven or more steps are treated as faults.

Run length	Runs	Median value ($\mu\text{g m}^{-3}$)	Share above $100 \mu\text{g m}^{-3}$
2	1736	65	20%
3	398	59	9%
4-5	145	61	10%
6-10	26	52	15%
11+	2	51	0%
whole record	n/a	74	29%

Table S34: **The paired comparison between feature sets.** The difference $\Delta_{\text{coordinate-free}} - \Delta_{\text{coordinate-bearing}}$ on the stations both specifications predict, under the ten-day guard. This is the comparison the manuscript reports, in place of contrasting each specification's separate verdict against zero, which is not a comparison of the two (Gelman and Stern, 2006). Positive favours the coordinate-free specification. The cell columns take the nine covariate-cell medians as the units.

<i>R</i> (km)	Stations	Sign <i>p</i>	Wilcoxon <i>p</i>	Cells	Cell sign <i>p</i>	Cell-block 95% CI
<i>distance buffer only</i>						
0	8/19	0.648	0.891	5/9	1.000	[-0.085, +0.140]
25	13/19	0.167	0.134	7/9	0.180	[-0.037, +0.390]
50	13/19	0.167	0.040	7/9	0.180	[-0.055, +0.500]
100	14/19	0.064	0.026	7/9	0.180	[+0.007, +0.997]
<i>buffer + time blocks</i>						
0	7/19	0.359	0.352	4/9	1.000	[-0.108, +0.046]
25	15/19	0.019	0.080	7/9	0.180	[-0.045, +0.148]
50	14/19	0.064	0.096	7/9	0.180	[-0.052, +0.514]
100	13/19	0.167	0.036	7/9	0.180	[-0.007, +0.601]
<i>buffer + time + leave-one-grid-cell-out</i>						
0	11/19	0.648	0.738	5/9	1.000	[-0.052, +0.206]
25	15/19	0.019	0.113	7/9	0.180	[-0.052, +0.148]
50	14/19	0.064	0.096	7/9	0.180	[-0.052, +0.514]
100	13/19	0.167	0.036	7/9	0.180	[-0.000, +0.601]

S19 Ground-truth stochastic experiment: full results

The tables in this section report the 32-scenario factorial experiment of Section 2.6 in full. Each cell aggregates 16 scenarios $\times$ 100 replicates = 1,600 realizations. Under $\beta = 0$ a positive transfer verdict is a false positive; under $\beta = 1$ it is a true detection. Verdict rates are percentages and $\tilde{\Delta}$ is the median across realizations of the per-realization median station-level $\Delta = \log(\text{MAE}_{\text{raw}}/\text{MAE}_{\text{transfer}})$.

Table S35: **The validation ladder under known ground truth, at all three pre-declared buffer radii.** Each rung adds one control to the rung above it. Every cell is 1,600 realizations. The false-positive rate falls by an order of magnitude between the buffer rung and the time-guard rung, and barely moves thereafter; detection falls at every rung and falls further for the coordinate-bearing specification as the buffer widens. At $R = 100$ km the grid-cell rung is exactly redundant, because the buffer has already removed every same-cell source.

R	Specification	$\beta = 0$: false-positive verdict rate (%)				$\beta = 1$: detection (%)	
		station	+buffer	+time	+cell	station	full audit
25	coordinate-free	20.00	16.69	2.50	2.31	77.94	36.81
	coordinate-bearing	29.50	17.56	3.69	2.94	79.56	27.81
50	coordinate-free	20.00	16.56	2.75	2.69	77.94	36.69
	coordinate-bearing	29.50	13.56	2.88	2.44	79.56	25.69
100	coordinate-free	20.00	15.56	2.62	2.62	77.94	35.31
	coordinate-bearing	29.50	10.62	2.06	2.06	79.56	22.50

Table S36: **Median transfer estimand $\tilde{\Delta}$ along the same ladder.** The verdict rule requires both $\tilde{\Delta} > 0$ and a one-sided sign test, so the estimand is reported separately from the verdict rate. Under $\beta = 0$ the full audit drives $\tilde{\Delta}$ increasingly negative as the buffer widens, which is the expected behaviour when a correction is transferred to a location it carries no information about.

R	Specification	$\beta = 0$				$\beta = 1$			
		station	+buffer	+time	+cell	station	+buffer	+time	+cell
25	coordinate-free	+0.0115	−0.0019	−0.1028	−0.1056	+0.2426	+0.2031	+0.0977	+0.0916
	coordinate-bearing	+0.0556	−0.0035	−0.0966	−0.1560	+0.2971	+0.2012	+0.0795	+0.0167
50	coordinate-free	+0.0115	−0.0050	−0.1100	−0.1123	+0.2426	+0.1914	+0.0811	+0.0801
	coordinate-bearing	+0.0556	−0.0301	−0.1549	−0.1873	+0.2971	+0.1335	+0.0178	−0.0228
100	coordinate-free	+0.0115	−0.0080	−0.1131	−0.1131	+0.2426	+0.1846	+0.0816	+0.0816
	coordinate-bearing	+0.0556	−0.1539	−0.2741	−0.2741	+0.2971	+0.0187	−0.1009	−0.1009

The headline specificity is a marginal average, and one corner of the factorial is not 2–3%

The full-audit false-positive rate of 2–3% reported in the main text is the rate averaged over the sixteen null scenarios. It is not uniform across them. In one corner of the factorial — temporal dependence present, spatial dependence absent, shared-grid dependence absent — the full audit still returns a positive transfer verdict in 11–13% of realizations, at every radius and for both feature sets (Table S40). Every other corner is at or below 4%.

The cause is not the guard, which is re-estimated at a median of three days in that corner, the same width as the adjacent corner that sits at 1–2%. Nor is it the estimand: the median Δ there

Table S37: **Which dependence pathway creates the false transfer verdict.** $\beta = 0$, $R = 50$ km, dense and sparse geometries pooled. Each row is one combination of the three simulated dependence pathways, so a row holds 200 realizations per specification and design: the two network geometries $\times$ 100 replicates, with β and the three pathway switches all fixed by the row. Station-only validation is harmless when no dependence is present and reports transfer in roughly half of the realizations once temporal dependence is switched on. The buffer alone does not repair this; the temporal guard does.

Dependence present			coordinate-free (%)				coordinate-bearing (%)			
spatial	temporal	grid	stat.	+buf	+time	+cell	stat.	+buf	+time	+cell
–	–	–	1.0	2.0	0.0	0.5	1.5	1.0	0.0	0.0
–	–	✓	6.0	4.0	1.0	1.5	18.0	3.5	0.5	0.0
–	✓	–	80.0	79.0	13.5	13.0	80.5	75.0	14.0	12.5
–	✓	✓	34.5	29.0	2.5	2.5	45.0	12.0	0.5	1.0
✓	–	–	1.5	0.0	0.0	0.0	13.0	3.0	2.0	2.5
✓	–	✓	1.0	0.5	0.0	0.0	13.5	3.0	2.5	1.5
✓	✓	–	19.5	9.0	2.5	1.5	34.5	4.5	2.0	1.0
✓	✓	✓	16.5	9.0	2.5	2.5	30.0	6.5	1.5	1.0

Table S38: **Network geometry changes what the grid-cell stage can still find.** The sparse geometry doubles source–target distances about the network centroid while holding grid-cell membership fixed, so it varies geometry without confounding the shared-grid factor. Each cell is 800 realizations (8 scenarios $\times$ 100 replicates).

R	Specification	Geometry	$\beta = 0$ false positive (%)		$\beta = 1$ detection (%)	
			+time	+cell	+time	+cell
25	coordinate-free	dense	2.9	2.5	36.2	35.5
		sparse	2.1	2.1	39.8	38.1
	coordinate-bearing	dense	3.8	3.6	34.2	27.8
		sparse	3.6	2.2	34.4	27.9
50	coordinate-free	dense	2.5	2.5	35.9	35.9
		sparse	3.0	2.9	38.8	37.5
	coordinate-bearing	dense	2.6	2.6	25.6	25.6
		sparse	3.1	2.2	30.4	25.8
100	coordinate-free	dense	2.9	2.9	34.4	34.4
		sparse	2.4	2.4	36.2	36.2
	coordinate-bearing	dense	2.0	2.0	21.4	21.4
		sparse	2.1	2.1	23.6	23.6

Table S39: **Data-adaptive guard width and surviving source count in the simulation.** The guard is re-estimated in every simulated source network by the same rule used on the observed data (first lag at which the median absolute source-station residual autocorrelation is at most 0.10, capped at 14 days), and never sees the target outcome. Median across realizations, with the interquartile range.

R	β	temporal dependence	guard (d), median [IQR]	surviving sources, median
25	0	off	1 [1, 2]	15.5
	0	on	3 [2, 5]	15.5
	1	off	4 [3, 5]	15.5
	1	on	4 [3, 6]	15.5
50	0	off	1 [1, 2]	14.5
	0	on	3 [2, 5]	14.5
	1	off	4 [3, 5]	14.5
	1	on	4 [3, 6]	14.5
100	0	off	1 [1, 2]	13.5
	0	on	3 [2, 5]	13.5
	1	off	4 [3, 5]	13.5
	1	on	4 [3, 6]	13.5

is -0.18 , as negative as anywhere else. The cause is the inference unit. When the only dependence present is a common temporal component, every station carries the same residual excursion, so the 19 station-level outcomes are near-perfectly dependent rather than independent, which is what the exact sign test across stations assumes. The distribution of the per-realization win rate in that corner has standard deviation 0.307 against 0.13–0.19 in every other corner, and is bimodal: 63% of realizations put fewer than a fifth of stations on the positive side and 9% put more than four fifths there, against 0% in the no-dependence corner. The mean win rate is in fact *lower* than in the no-dependence corner (0.228 versus 0.352); it is the inflated across-station variance, not a shift in central tendency, that produces the extra verdicts.

This is the same failure the main text indicts elsewhere in a different guise: a test is being read at an inference unit that the dependence structure does not support. It is reported here rather than smoothed into the average because an audit that conceals its own worst cell would be making the error the paper is about. It also identifies the condition under which a station-level sign test should not be the verdict rule — common-mode residual dependence with no spatial structure — and, by the same logic, supports the grid-cell block bootstrap used as the coarser inference unit in the empirical analysis.

Is the result a property of the transparent learner?

The simulation’s corrector is ridge regression on a fixed nonlinear basis, chosen so that the inferential weight falls on the validation design rather than on a model architecture. The empirical audit uses a histogram gradient-boosting regressor. A reader is entitled to ask whether the operating characteristics reported above survive that substitution, since a learner that partitions its own feature space can exploit a dependence pathway that a fixed basis cannot.

The experiment was therefore repeated with **HistGradientBoostingRegressor** under the settings the empirical audit uses (300 iterations, learning rate 0.05, maximum depth 6, internal early stopping disabled, `random_state = 0`), on the raw feature set rather than the radial basis: the mechanistic value, the seasonal pair, the date index, and, for the coordinate-bearing specification, the coordinates. The date index is given to the tree deliberately. Over a single simulated year it is a

Table S40: **Full-audit false-positive rate by dependence corner**, $\beta = 0$. Each cell is 200 realizations (2 geometries $\times$ 100 replicates). The corner with temporal dependence alone is boxed off from the rest by an order of magnitude and is stable across all three radii.

Dependence present			$R = 25$		$R = 50$		$R = 100$	
spatial	temporal	grid	c-free	c-bear	c-free	c-bear	c-free	c-bear
–	–	–	0.0	0.0	0.5	0.0	0.5	0.0
–	–	✓	1.5	1.5	1.5	0.0	2.5	1.5
–	✓	–	12.5	11.5	13.0	12.5	12.0	11.0
–	✓	✓	1.5	0.5	2.5	1.0	2.5	1.0
✓	–	–	0.0	4.0	0.0	2.5	0.0	2.0
✓	–	✓	0.5	2.5	0.0	1.5	0.5	1.0
✓	✓	–	1.0	2.0	1.5	1.0	2.0	0.0
✓	✓	✓	1.5	1.5	2.5	1.0	1.0	0.0

bijection onto the day, which is the pathway the temporal guard exists to close, and withholding it would make the tree easier to audit than the estimator used on the real network. Twenty-five replicates per scenario were run rather than one hundred, because a gradient-boosted fit is roughly two orders of magnitude more expensive than a ridge solve and the experiment refits the model once per target, design, specification and replicate. Each aggregated cell is therefore 400 realizations rather than 1,600, and the intervals below are correspondingly wider. The complete ladder was run at the primary radius. At the two sensitivity radii only the full-audit rung was run, because the proposition under test there is whether the operating characteristics of the *audited* design depend on the corrector, and the intermediate rungs at those radii do not bear on it. Before relying on that restriction we checked it: a run with the design loop restricted returns rows bit-identical to the same rows of an unrestricted run.

Comparing two learners means testing the difference between them, not asking whether their separate intervals overlap. That is the distinction this manuscript insists on elsewhere for the two feature sets (Gelman and Stern, 2006), and it applies here to us. Each pair is therefore tested directly, by Fisher’s exact test on the underlying verdict counts, with Benjamini–Hochberg applied across the twelve full-audit comparisons of Table S42. The distinction is not decorative: the interval reading declares all twelve pairs identical, and the direct test does not.

False-positive control is learner-invariant at every radius. Under $\beta = 0$ the full audit returns 1.00–1.75% under the tree against 2.06–2.94% under ridge across the three radii. None of the six pairs differs (p from 0.061 to 0.440; every $q > 0.35$), and the tree sits on the conservative side of all six. Specificity also improves monotonically as the buffer widens under the tree — 1.50, 1.25, 1.00% coordinate-free and 1.75, 1.25, 1.25% coordinate-bearing at 25, 50 and 100 km — which is the direction the design predicts and the first of the three pre-declared sanity conditions.

Retained detection is learner-invariant except at one cell, and there the tree is the more optimistic learner. Eleven of the twelve comparisons return $p \geq 0.05$. The exception is coordinate-bearing detection at $R = 100$ km: 28.75% under the tree against 22.50% under ridge, a difference of +6.25 percentage points [+1.37, +11.14], $p = 0.010$, which does not survive the correction across the family ($q = 0.12$). We report it rather than let the interval overlap absorb it, because it qualifies a secondary claim. The decline of coordinate-bearing detection as the buffer widens — 27.81% at 25 km to 22.50% at 100 km under ridge, $p = 0.001$ — is not reproduced under the tree, which moves from 30.75% to 28.75% ($p = 0.59$) on a quarter of the sample. The specificity result carries across both learners and all three radii. The size of the coordinate-bearing power loss at the widest buffer is a property the two learners do not share, and the main text states it as a

ridge-based result.

The unaudited rungs are not learner-invariant, and the tree is worse. At the primary radius, the only one at which the intermediate rungs were run, a plain station holdout makes the tree declare transfer in 52.0% and 56.3% of null realizations against 20.0% and 29.5% under ridge: +32.0 percentage points [+26.7, +37.3] and +26.8 [+21.4, +32.1], both with $p < 10^{-22}$ by the same direct test. The distance buffer alone leaves the gap open (28.3% and 24.0% against 16.6% and 13.6%; $p < 10^{-6}$ for both). Only after the temporal guard do the two learners stop differing ($p = 0.29$ and $p = 0.075$).

The consequence for the empirical arm is not neutral. The real network is audited with the tree, so the ridge-based rates reported in the main text *understate* how misleading an unaudited station holdout would be in that setting. A flexible learner does not weaken the case for controlling the dependence pathways; it is the case.

Table S41: **Learner sensitivity of the simulated audit, $R = 50$ km.** Verdict rates in per cent with Wilson 95% intervals. Ridge cells are 1,600 realizations (16 scenarios $\times$ 100 replicates); gradient-boosting cells are 400 (16 $\times$ 25). $\tilde{\Delta}$ is the median across realizations of the per-realization median station-level Δ .

β	Specification	Design	Ridge (RBF basis)	Gradient boosting	
			rate [95% CI]	rate [95% CI]	$\tilde{\Delta}$
0	coordinate-free	station	20.00 [18.11, 22.03]	52.00 [47.11, 56.85]	+0.1316
		+ buffer	16.56 [14.82, 18.46]	28.25 [24.06, 32.85]	−0.0115
		+ time guard	2.75 [2.05, 3.67]	1.75 [0.85, 3.57]	−0.1098
		+ grid cell	2.69 [2.00, 3.60]	1.25 [0.54, 2.89]	−0.1103
	coordinate-bearing	station	29.50 [27.32, 31.78]	56.25 [51.35, 61.03]	+0.2360
		+ buffer	13.56 [11.97, 15.33]	24.00 [20.07, 28.42]	−0.0203
		+ time guard	2.88 [2.16, 3.81]	1.25 [0.54, 2.89]	−0.0912
		+ grid cell	2.44 [1.79, 3.31]	1.25 [0.54, 2.89]	−0.0958
1	coordinate-free	station	77.94 [75.84, 79.90]	93.50 [90.65, 95.53]	+0.3534
		+ buffer	63.38 [60.98, 65.70]	64.00 [59.18, 68.55]	+0.2405
		+ time guard	37.31 [34.98, 39.71]	35.25 [30.73, 40.05]	+0.0750
		+ grid cell	36.69 [34.36, 39.08]	34.75 [30.25, 39.54]	+0.0672
	coordinate-bearing	station	79.56 [77.52, 81.47]	92.75 [89.78, 94.90]	+0.4257
		+ buffer	43.56 [41.15, 46.00]	54.00 [49.10, 58.82]	+0.2258
		+ time guard	28.00 [25.85, 30.25]	30.75 [26.43, 35.44]	+0.0595
		+ grid cell	25.69 [23.61, 27.88]	28.75 [24.53, 33.37]	+0.0502

Does the simulated ladder span the same deployment geometry as the real one?

The simulation carries the paper’s generality claim, so it has to be shown that its validation ladder reaches the same deployment geometry as the observed network’s. Tables S27 and S28 report the nearest-neighbour distance structure and the area of applicability for the real audit; the same two diagnostics are computed here for the simulated networks by `simulation/aoa_knndm_sim.py`.

The distance structure is the quantity kNNDM minimises (Milà et al., 2022; Linnenbrink et al., 2024): the discrepancy between the distance from a target to its nearest surviving source and the nearest-neighbour distance *within* the surviving source set. The dissimilarity index and its threshold follow Meyer and Pebesma (2021), in the unweighted variant — the feature-importance weighting of the original formulation needs a fitted model, and what is at issue here is the design rather than

Table S42: **Learner sensitivity of the full audit across all three pre-declared radii.** Verdict rates in per cent for the full audit (distance buffer + time guard + grid-cell exclusion). Ridge cells are 1,600 realizations (16 scenarios $\times$ 100 replicates); gradient-boosting cells are 400 (16 $\times$ 25). The comparison is the *paired difference between learners*, tested directly by Fisher’s exact test, not by asking whether the two intervals overlap: the overlap reading declares all twelve pairs identical, and one of them is not. q is Benjamini–Hochberg across this family of twelve.

β	Specification	R (km)	Ridge (%)	Tree (%)	Difference [95% CI]	p	(q)
0	coordinate-free	25	2.31	1.50	−0.81 [−2.21, +0.59]	0.440	(0.58)
		50	2.69	1.25	−1.44 [−2.78, −0.09]	0.102	(0.41)
		100	2.62	1.00	−1.62 [−2.88, −0.37]	0.061	(0.36)
0	coordinate-bearing	25	2.94	1.75	−1.19 [−2.72, +0.34]	0.229	(0.45)
		50	2.44	1.25	−1.19 [−2.51, +0.14]	0.183	(0.45)
		100	2.06	1.25	−0.81 [−2.10, +0.48]	0.411	(0.58)
1	coordinate-free	25	36.81	36.25	−0.56 [−5.83, +4.71]	0.862	(0.91)
		50	36.69	34.75	−1.94 [−7.17, +3.29]	0.486	(0.58)
		100	35.31	35.75	+0.44 [−4.81, +5.69]	0.907	(0.91)
1	coordinate-bearing	25	27.81	30.75	+2.94 [−2.09, +7.96]	0.264	(0.45)
		50	25.69	28.75	+3.06 [−1.86, +7.99]	0.228	(0.45)
		100	22.50	28.75	+6.25 [+1.37, +11.13]	0.010 *	(0.12)

* raw $p < 0.05$; does not survive Benjamini–Hochberg across the twelve comparisons.

one corrector. Station geometry, cell assignment and the feature matrix are deterministic given the geometry factor and the seed, so one seeded realization per cell suffices.

Three things follow.

Plain leave-one-station-out validates interpolation in the simulation too. At zero buffer the target-to-source and within-source distance distributions coincide (both medians 6.55 km in the dense geometry, 13.10 km in the sparse one; $W_1 = 1.98$ and 3.98 km). This is the same statement Table S27 makes about the observed network, and it is what makes the station rung’s 20–29% false-positive rate interpretable rather than surprising.

The ladder reaches genuine extrapolation. Under the full audit the median target-to-source distance rises to 107.8 km in the dense geometry and 147.0 km in the sparse one, against a within-source nearest neighbour that stays near 7 and 14 km; W_1 reaches 86.4 and 103.0 km.

The coordinate result reproduces independently. In the observed network the coordinate-bearing specification at a 100 km buffer is inadmissible everywhere by the area-of-applicability criterion (median DI 6.76, 100% of points outside) while the coordinate-free one stays admissible (DI 1.23, 10.5% outside). The simulated networks reproduce the structure and the order of magnitude without being tuned to do so: coordinate-bearing DI rises from 1.44 to 4.19 across the buffer ladder with 66.4% of points outside at 100 km, whereas coordinate-free DI stays between 0.81 and 0.88 with at most 9.6% outside. The empirical and the simulated arms therefore agree on the mechanism, not only on the verdict: coordinates do not become harmful, they become inadmissible, and the buffer is what exposes it.

Table S43: **Distance structure and area of applicability of the simulated validation ladder.** Station rung at zero buffer, and the full audit at each pre-declared radius, for both simulated geometries. Distances in km; W_1 is the Wasserstein-1 distance between the target-to-source and within-source nearest-neighbour distributions, which is what kNNDM minimises. DI is the un-weighted dissimilarity index and the last column the share of target prediction points above the area-of-applicability threshold. Compare Tables S27 and S28 for the observed network.

Geometry	Specification	Rung	R	target→source	within source	W_1	DI [outside]
dense	coordinate-free	station	0	6.5	6.5	2.0	0.59 [1.2%]
		full audit	0	48.3	6.5	24.6	0.86 [7.9%]
		full audit	25	62.6	7.0	37.4	0.88 [9.6%]
		full audit	50	73.5	7.0	51.4	0.86 [8.5%]
		full audit	100	107.8	7.0	86.4	0.81 [9.1%]
dense	coordinate-bearing	station	0	6.5	6.5	2.0	0.70 [1.5%]
		full audit	0	48.3	6.5	24.6	1.44 [17.5%]
		full audit	25	62.6	7.0	37.4	1.78 [17.5%]
		full audit	50	73.5	7.0	51.4	2.18 [33.9%]
		full audit	100	107.8	7.0	86.4	4.19 [66.4%]
sparse	coordinate-free	station	0	13.1	13.1	4.0	0.59 [1.2%]
		full audit	0	96.6	13.1	49.2	0.86 [7.9%]
		full audit	25	96.6	13.6	51.2	0.86 [8.5%]
		full audit	50	125.1	14.0	74.8	0.88 [9.6%]
		full audit	100	147.0	14.0	102.9	0.86 [8.5%]
sparse	coordinate-bearing	station	0	13.1	13.1	4.0	0.70 [1.5%]
		full audit	0	96.6	13.1	49.2	1.44 [17.5%]
		full audit	25	96.6	13.6	51.2	1.44 [17.5%]
		full audit	50	125.1	14.0	74.8	1.78 [17.5%]
		full audit	100	147.0	14.0	102.9	2.18 [33.9%]